

Theoretical Background: SAF Skyrmion Simulator

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Contents

1	Magnetic Skyrmions in Synthetic Antiferromagnets	2
1.1	Notation	2
1.2	Units	3
2	Hamiltonian	3
2.1	Exchange Energy	3
2.2	Dzyaloshinskii–Moriya Interaction Energy	4
2.3	Anisotropy Energy	4
2.4	Demagnetization Energy	5
2.5	Zeeman Energy	6
2.6	RKKY Interlayer Coupling Energy	7
2.7	Total Energy on the Discrete Lattice	7
3	From Hamiltonian to Effective Field	7
3.1	Variational Derivative	7
3.2	Derivation of the Exchange Field	7
3.3	Derivation of the DMI Field	8
3.4	Derivation of the Anisotropy Field	8
3.5	Derivation of the Zeeman and RKKY Fields	8
3.6	Physical Interpretation of the LLG Equation	9
4	Effective Field Contributions	9
4.1	Exchange Field	9
4.2	Dzyaloshinskii–Moriya Interaction	9
4.3	Anisotropy and demagnetization	10
4.3.1	Perpendicular magnetic anisotropy: discrete form	10
4.3.2	PMA vs demagnetization: same form, different physics	11
4.3.3	Demagnetization kernels	13
4.4	Zeeman Field	16
4.5	RKKY Interlayer Coupling	16
5	Landau–Lifshitz–Gilbert–Slonczewski Equation (LLGS)	17
5.1	Explicit Form	17
5.2	Thermal Fluctuations	19

6	Spin-Orbit Torques	19
6.1	Spin Hall Effect	19
6.2	SOT Torques in the LLGS Equation	19
6.3	SOT Terms in the Explicit LLGS	20
6.4	Topological Spin Hall Torque	20
7	Magnus Force and the Thiele Equation	22
8	Topological Charge	23
8.1	Discretization	24
9	Skyrmion Profile (Initial Condition)	25
10	Simulation Parameters	25
11	Implementation Notes	26
11.1	Local $N_z = 1$ demag vs full FFT demag	26
11.2	DMI uncertainty: $D \pm 39\%$	27
11.3	Anisotropy uncertainty: $K \pm 7\%$	27
11.4	Gilbert damping uncertainty: $\alpha \pm 0.04$	27
11.5	Summary	28
11.6	Time Integration	28
11.7	Boundary Conditions	28
11.8	Output Format	28
12	Relaxation Convergence and Finite-Size Independence	29
12.1	Free top/bottom boundaries: the $D > D_c$ expansion	30
12.2	Boundary conditions: periodic vs racetrack vs masked band	32
12.3	DMI sweep: dialing a stable, larger racetrack skyrmion	33
A	Derivation of the Domain Wall Profile	35
B	The Constraint $\mathbf{m} \cdot d\mathbf{m}/dt = 0$	36

1 Magnetic Skyrmions in Synthetic Antiferromagnets

Magnetic skyrmions are topologically non-trivial chiral spin textures with a topological charge $S = \pm 1$. In a skyrmion, the magnetization rotates smoothly from one out-of-plane direction at the core to the opposite direction in the surrounding domain.

A synthetic antiferromagnet (SAF) consists of two ferromagnetic layers (Co) separated by a non-magnetic spacer (Ru), coupled antiferromagnetically via the Ruderman–Kittel–Kasuya–Yosida (RKKY) interaction. In a SAF, skyrmions in the two layers have opposite core polarities and opposite topological charges, leading to a cancellation of the Magnus force and a vanishing skyrmion Hall effect [1].

The SAF stack modeled here is Pt/Co/Ru/Pt/Co/Ru, following the experimental system in [1].

1.1 Notation

- $\mathbf{M}(\mathbf{r}, t)$ — magnetization.
- M_s — saturation magnetization at the operating temperature, held fixed.

- $\mathbf{m} = \mathbf{M}/M_s$ — unit magnetization vector, $|\mathbf{m}| = 1$; evolved by the LLGS equation.
- $\mathbf{H}_{\text{eff}}, \mathbf{H}_{\text{ex}}, \mathbf{H}_{\text{DMI}}, \mathbf{H}_{\text{anis}}, \mathbf{H}_{\text{demag}}, \mathbf{H}_{\text{RKKY}}, \mathbf{H}_{\text{ext}}$ — effective fields entering the LLGS equation $d\mathbf{m}/dt = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}}$. The $\mathbf{B} = \mu_0 \mathbf{H}$ factor is absorbed into the prefactors (B-field units; see Sec. 1.2).
- J_{RKKY} — RKKY interlayer coupling constant; surface energy density (J/m^2) in Section 2.6, volume energy density (J/m^3) in Sec. 3, related by $J_{\text{RKKY}}^{(\text{vol})} = J_{\text{RKKY}}^{(\text{surf})}/t_{\text{Co}}$.
- H_{RKKY} — RKKY field magnitude: $H_{\text{RKKY}} = J_{\text{RKKY}}^{(\text{vol})}/M_s = J_{\text{RKKY}}^{(\text{surf})}/(M_s t_{\text{Co}})$.
- $\mathbf{J} = J(t) \hat{\mathbf{J}}$ — current density; $J(t)$ is the magnitude, prescribed as a current pulse (a function of time), and $\hat{\mathbf{J}}$ the unit vector along the current flow.
- $\hat{\mathbf{p}}$ — spin polarization direction, $\hat{\mathbf{p}} = \hat{\mathbf{z}} \times \hat{\mathbf{J}}$.
- $\hat{\mathbf{z}}$ — film-normal unit vector (out-of-plane).
- $H_{\text{DL}} = \chi_{\text{DL}} J(t)$, $H_{\text{FL}} = \chi_{\text{FL}} J(t)$ — damping-like and field-like SOT effective fields.
- $K_{\text{eff}} = K - \mu_0 M_s^2/2$ — effective uniaxial anisotropy with thin-film shape correction.
- $V_{\text{cell}} = t_{\text{Co}} a^2$ — per-site simulation volume.
- Energy integrals over a layer: $\int \cdots dV = t_{\text{Co}} \int \cdots dS$ for thin uniform layers.

1.2 Units

All quantities are in SI. Effective fields are carried in Tesla (B-field units), absorbing the μ_0 factor from $\mathbf{B} = \mu_0 \mathbf{H}$.

2 Hamiltonian

The total magnetic energy of the SAF system is the sum of contributions from both layers ($\ell = \text{top, bot}$) plus the interlayer coupling:

$$\mathcal{H} = \sum_{\ell} \left(\mathcal{H}_{\text{ex}}^{(\ell)} + \mathcal{H}_{\text{DMI}}^{(\ell)} + \mathcal{H}_{\text{anis}}^{(\ell)} + \mathcal{H}_{\text{demag}}^{(\ell)} + \mathcal{H}_Z^{(\ell)} \right) + \mathcal{H}_{\text{RKKY}} \quad (1)$$

The effective field entering the LLGS equation is obtained from the variational derivative of the energy with respect to the magnetization:

$$\mathbf{H}_{\text{eff}} = -\frac{1}{M_s} \frac{\delta \mathcal{H}}{\delta \mathbf{m}} \quad (2)$$

2.1 Exchange Energy

The Heisenberg exchange favors parallel alignment of neighboring spins:

$$\mathcal{H}_{\text{ex}} = A_{\text{ex}} \int [(\nabla m_x)^2 + (\nabla m_y)^2 + (\nabla m_z)^2] dV \quad (3)$$

On the discrete 2D lattice with lattice constant a and film thickness t_{Co} :

$$\mathcal{H}_{\text{ex}} = -2A_{\text{ex}} t_{\text{Co}} \sum_{\langle i,j \rangle} \mathbf{m}_i \cdot \mathbf{m}_j \quad (4)$$

where the sum runs over nearest-neighbor pairs, each counted once (constant terms dropped). Expanding $|\nabla \mathbf{m}|^2 \approx \sum_{\delta} |\mathbf{m}_{i+\delta} - \mathbf{m}_i|^2/(2a^2)$ and using $|\mathbf{m}| = 1$ gives the factor 2.

Symbol	Quantity	Unit
$\mathbf{M}, M_s$	magnetization, saturation magnetization	A/m
$\mathbf{m}$	unit magnetization vector	—
$d\mathbf{m}/dt$	magnetization rate of change	s ⁻¹
$\mathbf{H}_{\text{eff}}, \mathbf{H}_{\text{ex}}, \mathbf{H}_{\text{DMI}}, \mathbf{H}_{\text{anis}}$	effective fields	T
$\mathbf{H}_{\text{demag}}, \mathbf{H}_{\text{RKKY}}, \mathbf{H}_{\text{ext}}, \mathbf{B}_{\text{ext}}$	effective fields	T
$H_{\text{DL}}, H_{\text{FL}}$	SOT effective fields	T
A_{ex}	exchange stiffness	J/m
D	DMI constant	J/m ²
$K, K_{\text{top}}, K_{\text{bot}}, K_{\text{eff}}$	uniaxial anisotropy	J/m ³
α	Gilbert damping	—
γ	gyromagnetic ratio	rad/(s·T)
μ_0	vacuum permeability	T·m/A
$J_{\text{RKKY}}^{(\text{surf})}$	RKKY surface energy density	J/m ²
$J_{\text{RKKY}}^{(\text{vol})}$	RKKY volume energy density	J/m ³
H_{RKKY}	RKKY field magnitude	T
$\mathbf{J}, J$	current density, magnitude	A/m ²
$\hat{J}, \hat{\mathbf{p}}, \hat{z}$	unit vectors	—
$\chi_{\text{DL}}, \chi_{\text{FL}}$	SOT coefficients	T·A ⁻¹ ·m ²
a	lattice constant	m
$t_{\text{Co}}, d_{\text{Ru}}$	layer / spacer thicknesses	m
Δ	domain wall width	m
R	skyrmion radius	m
$V_{\text{cell}} = t_{\text{Co}} a^2$	per-site simulation volume	m ³
n_x, n_y	lattice sites	—
$t, \Delta t$	time, time step	s
$\mathcal{H}$	Hamiltonian (energy)	J
Q	topological charge	—
$\mathbf{G}$	gyrovector	—

2.2 Dzyaloshinskii–Moriya Interaction Energy

The interfacial (Néel-type) DMI at the Pt/Co interface introduces chirality:

$$\mathcal{H}_{\text{DMI}} = D t_{\text{Co}} \int [m_z (\nabla \cdot \mathbf{m}) - (\mathbf{m} \cdot \nabla) m_z] dS \quad (5)$$

On the discrete lattice this becomes a sum over nearest-neighbor bonds with bond-direction-dependent cross products:

$$\mathcal{H}_{\text{DMI}} = -D t_{\text{Co}} a \sum_{\langle i,j \rangle} (\hat{r}_{ij} \times \hat{z}) \cdot (\mathbf{m}_i \times \mathbf{m}_j) \quad (6)$$

where $\hat{r}_{ij}$ is the unit vector from site i to site j and each bond is counted once. The factor a restores energy units: each bond carries a gradient $\sim 1/a$ over a cell area a^2 .

2.3 Anisotropy Energy

Uniaxial perpendicular magnetic anisotropy (PMA) along the film normal:

$$\mathcal{H}_{\text{anis}} = -K t_{\text{Co}} \int m_z^2 dS \quad (7)$$

On the lattice:

$$\mathcal{H}_{\text{anis}} = -K t_{\text{Co}} a^2 \sum_i m_{z,i}^2 \quad (8)$$

PMA in Co/Pt and Co/Ru/Pt stacks originates from interfacial spin-orbit coupling at the heavy-metal interface (3d–5d hybridization between Co and Pt). The effect scales with the interface area, so thinner Co (here $t_{\text{Co}} = 1.3$ nm) has a larger interface-to-volume ratio and stronger PMA per unit volume.

2.4 Demagnetization Energy

In the thin-film limit ($t_{\text{Co}} \ll$ lateral dimensions), the magnetostatic energy reduces to a local shape anisotropy:

$$\mathcal{H}_{\text{demag}} = \frac{\mu_0 M_s^2}{2} t_{\text{Co}} \int m_z^2 dS \quad (9)$$

This opposes out-of-plane magnetization. Since both the anisotropy and demagnetization energies are proportional to m_z^2 , they merge into a single effective anisotropy:

$$\mathcal{H}_{\text{anis}} + \mathcal{H}_{\text{demag}} = - \underbrace{\left(K - \frac{\mu_0 M_s^2}{2} \right)}_{K_{\text{eff}}} t_{\text{Co}} \int m_z^2 dS \quad (10)$$

Three regimes arise depending on the sign of K_{eff} :

- $K > \mu_0 M_s^2/2$: $K_{\text{eff}} > 0$ — anisotropy wins, magnetization points out-of-plane.
- $K < \mu_0 M_s^2/2$: $K_{\text{eff}} < 0$ — demagnetization wins, magnetization lies in-plane.
- $K \approx \mu_0 M_s^2/2$: $K_{\text{eff}} \approx 0$ — spin reorientation transition (SRT); domain walls and skyrmions cost very little energy to form.

For the Co/Pt system in [1], $\mu_0 M_s^2/2 \approx 1.285 \times 10^6$ J/m³. The measured top-layer anisotropy $K \approx 1.294 \times 10^6$ J/m³ (anisotropy field $\mu_0 H_k = 12.4$ mT) gives $K_{\text{eff}} \approx 9 \times 10^3$ J/m³; the value used in the paper’s simulations and in this simulator, $K_{\text{top}} = 1.3106 \times 10^6$ J/m³ ($\mu_0 H_k = 36$ mT), gives $K_{\text{eff}} \approx 2.6 \times 10^4$ J/m³. Either way K_{eff} is barely positive, right at the SRT, and this is why the system can host skyrmions.

Interface PMA (Perpendicular Magnetic Anisotropy) vs shape anisotropy: the spin reorientation transition

In a thin ferromagnetic film, three anisotropy sources coexist:

1. **Volume (bulk crystal) anisotropy** (K_{vol} , J/m³) — intrinsic magnetocrystalline plus magneto-elastic anisotropy of the Co bulk, per unit volume, independent of layer thickness. It is grouped together with the interfacial $2K_s/t$ into the single bare anisotropy $K = K_{\text{vol}} + 2K_s/t$ carried by the simulator. Sputter-deposited Co/Pt multilayers give $K_{\text{vol}} \approx -1.0 \times 10^6$ J/m³ [7].
2. **Interfacial PMA** (K_s , J/m²) — created by spin-orbit hybridization at heavy-metal interfaces (Pt, Ir, W, Pd, Au). Energy per unit *interface area*; spread over the layer volume each interface contributes K_s/t per unit volume, so two equivalent interfaces give $2K_s/t$. Favors out-of-plane.
3. **Shape anisotropy** ($\frac{1}{2}\mu_0 M_s^2$, J/m³) — from the demag field of a thin slab uniformly magnetized along $\hat{z}$. Per unit volume, independent of t . Favors in-plane.

The per-volume effective anisotropy of a film of thickness t with two equivalent interfaces is

$$K_{\text{eff}} = \underbrace{K_{\text{vol}} + \frac{2K_s}{t}}_K - \frac{1}{2}\mu_0 M_s^2, \quad (11)$$

where:

- K_s/t scales as $1/t$ — interfaces dominate as t shrinks.
- $\frac{1}{2}\mu_0 M_s^2$ is independent of thickness — pure volume term.

Sign convention:

- $K_{\text{eff}} > 0 \implies$ out-of-plane easy axis (PMA).
- $K_{\text{eff}} < 0 \implies$ in-plane easy axis.

There is a critical thickness t_c where K_{eff} changes sign:

$$t_c = \frac{2K_s}{\frac{1}{2}\mu_0 M_s^2 - K_{\text{vol}}}. \quad (12)$$

For Co/Pt: $K_s \sim 5 \times 10^{-4} - 10^{-3} \text{ J/m}^2$, $\frac{1}{2}\mu_0 M_s^2 \sim 1.3 \times 10^6 \text{ J/m}^3$, K_{vol} tiny $\implies t_c \sim 1 \text{ nm}$.

- $t < t_c$: PMA wins, magnetization out-of-plane.
- $t > t_c$: shape anisotropy wins, magnetization in-plane.

The simulator's Co layer is $t_{\text{Co}} = 1.3 \text{ nm}$ — **right at the SRT, deliberately**. That is why $K_{\text{eff}} = K_{\text{top}} - \mu_0 M_s^2/2 \approx 2.6 \times 10^4 \text{ J/m}^3$ is a small difference of two large numbers ($\sim 10^6$ each), and why a 7% uncertainty in K exceeds K_{eff} itself (see Implementation Notes, Sec. 11).

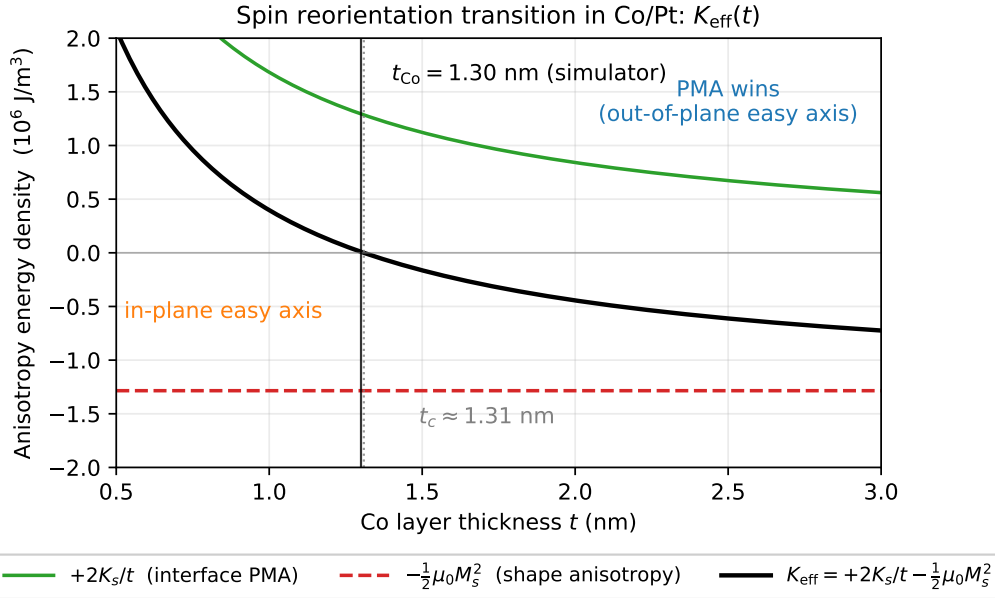


Figure 1: Spin reorientation transition (SRT) in Co/Pt. The interface PMA $+2K_s/t$ (green) decreases with the Co thickness t ; the shape anisotropy $-\frac{1}{2}\mu_0 M_s^2$ (red, dashed) is thickness-independent. Their sum K_{eff} (black) crosses zero at the critical thickness t_c , where the easy axis flips from out-of-plane (PMA wins, $K_{\text{eff}} > 0$, blue region) to in-plane (shape wins, $K_{\text{eff}} < 0$, orange region). With K_s set so that $K_{\text{top}} = 2K_s/t_{\text{Co}}$ matches the simulator's bare anisotropy ($K_s \approx 8.5 \times 10^{-4} \text{ J/m}^2$), $t_c \approx t_{\text{Co}} = 1.3 \text{ nm}$. The Co layer of [1] sits exactly at the SRT — which is why K_{eff} is the small difference of two $\sim 10^6 \text{ J/m}^3$ numbers.

2.5 Zeeman Energy

Interaction with an external field $\mathbf{B}_{\text{ext}}$:

$$\mathcal{H}_Z = -M_s t_{\text{Co}} \int \mathbf{m} \cdot \mathbf{B}_{\text{ext}} dS \quad (13)$$

2.6 RKKY Interlayer Coupling Energy

The antiferromagnetic RKKY exchange between the two Co layers across the Ru spacer:

$$\mathcal{H}_{\text{RKKY}} = J_{\text{RKKY}} \int \mathbf{m}_{\text{top}} \cdot \mathbf{m}_{\text{bot}} dS \quad (14)$$

where $J_{\text{RKKY}} > 0$ for antiferromagnetic coupling (surface energy density, J/m²). In the Tesla convention adopted here, the coupling field magnitude is $H_{\text{RKKY}} = J_{\text{RKKY}}/(M_s t_{\text{Co}})$ (cf. Notation block).

2.7 Total Energy on the Discrete Lattice

Combining all contributions, the total energy per layer on the 2D lattice is:

$$\mathcal{H}^{(\ell)} = t_{\text{Co}} a^2 \sum_i \left[-\frac{A_{\text{ex}}}{a^2} \sum_{\delta} \mathbf{m}_i \cdot \mathbf{m}_{i+\delta} - \frac{D}{2a} \sum_{\delta} (\hat{\delta} \times \hat{z}) \cdot (\mathbf{m}_i \times \mathbf{m}_{i+\delta}) - K_{\text{eff}} m_{z,i}^2 - M_s \mathbf{m}_i \cdot \mathbf{B}_{\text{ext}} \right] \quad (15)$$

where δ runs over the four nearest neighbors $\{\pm\hat{x}, \pm\hat{y}\}$. The site sum visits every bond twice, which is why the exchange and DMI coefficients here are half of the per-bond values above. The interlayer RKKY term adds $J_{\text{RKKY}} a^2 \sum_i \mathbf{m}_{\text{top},i} \cdot \mathbf{m}_{\text{bot},i}$.

3 From Hamiltonian to Effective Field

The Hamiltonian (Section 2) defines *what* the system wants to minimize. The LLGS equation (Section 5) describes *how* the magnetization evolves in time. The bridge between the two is the **effective field**, obtained from the variational derivative of the energy with respect to the magnetization.

3.1 Variational Derivative

Consider a small perturbation $\mathbf{m} \rightarrow \mathbf{m} + \delta\mathbf{m}$ (subject to the constraint $|\mathbf{m}| = 1$, so $\mathbf{m} \cdot \delta\mathbf{m} = 0$). The corresponding change in energy is:

$$\delta\mathcal{H} = \int \frac{\delta\mathcal{H}}{\delta\mathbf{m}} \cdot \delta\mathbf{m} dV \quad (16)$$

The effective magnetic field is defined such that:

$$\boxed{\mathbf{H}_{\text{eff}} = -\frac{1}{M_s} \frac{\delta\mathcal{H}}{\delta\mathbf{m}}} \quad (17)$$

so that $\delta\mathcal{H} = -M_s \int \mathbf{H}_{\text{eff}} \cdot \delta\mathbf{m} dV$. The factor $1/M_s$ gives $\mathbf{H}_{\text{eff}}$ the dimensions of magnetic field (Tesla in our convention).

3.2 Derivation of the Exchange Field

Starting from the exchange energy:

$$\mathcal{H}_{\text{ex}} = A_{\text{ex}} \int |\nabla\mathbf{m}|^2 dV \quad (18)$$

Perturbing $\mathbf{m} \rightarrow \mathbf{m} + \delta\mathbf{m}$:

$$\delta\mathcal{H}_{\text{ex}} = 2A_{\text{ex}} \int \nabla\mathbf{m} : \nabla(\delta\mathbf{m}) dV \quad (19)$$

where the colon denotes the tensor double contraction $\nabla \mathbf{m} : \nabla(\delta \mathbf{m}) = \sum_{\alpha} \nabla m_{\alpha} \cdot \nabla(\delta m_{\alpha})$. Integrating by parts (boundary terms vanish for periodic boundary conditions):

$$\delta \mathcal{H}_{\text{ex}} = -2A_{\text{ex}} \int \nabla^2 \mathbf{m} \cdot \delta \mathbf{m} dV \quad (20)$$

Reading off the variational derivative and applying (17):

$$\frac{\delta \mathcal{H}_{\text{ex}}}{\delta \mathbf{m}} = -2A_{\text{ex}} \nabla^2 \mathbf{m} \quad \implies \quad \mathbf{H}_{\text{ex}} = \frac{2A_{\text{ex}}}{M_s} \nabla^2 \mathbf{m} \quad (21)$$

3.3 Derivation of the DMI Field

From the interfacial DMI energy:

$$\mathcal{H}_{\text{DMI}} = D \int [m_z(\nabla \cdot \mathbf{m}) - (\mathbf{m} \cdot \nabla)m_z] dV \quad (22)$$

Expanding each term under the variation $\mathbf{m} \rightarrow \mathbf{m} + \delta \mathbf{m}$ and collecting contributions from δm_x , δm_y , and δm_z separately (using integration by parts where needed):

$$\frac{\delta \mathcal{H}_{\text{DMI}}}{\delta m_x} = -2D \frac{\partial m_z}{\partial x}, \quad \frac{\delta \mathcal{H}_{\text{DMI}}}{\delta m_y} = -2D \frac{\partial m_z}{\partial y}, \quad \frac{\delta \mathcal{H}_{\text{DMI}}}{\delta m_z} = 2D \left(\frac{\partial m_x}{\partial x} + \frac{\partial m_y}{\partial y} \right) \quad (23)$$

Applying (17):

$$\mathbf{H}_{\text{DMI}} = \frac{2D}{M_s} \left(\frac{\partial m_z}{\partial x} \hat{x} + \frac{\partial m_z}{\partial y} \hat{y} - \left(\frac{\partial m_x}{\partial x} + \frac{\partial m_y}{\partial y} \right) \hat{z} \right) \quad (24)$$

3.4 Derivation of the Anisotropy Field

From the anisotropy energy $\mathcal{H}_{\text{anis}} = -K \int m_z^2 dV$:

$$\delta \mathcal{H}_{\text{anis}} = -2K \int m_z \delta m_z dV = -2K \int (m_z \hat{z}) \cdot \delta \mathbf{m} dV \quad (25)$$

Therefore:

$$\mathbf{H}_{\text{anis}} = \frac{2K}{M_s} m_z \hat{z} \quad (26)$$

The demagnetization field follows identically with $K \rightarrow -\mu_0 M_s^2/2$, giving $\mathbf{H}_{\text{demag}} = -\mu_0 M_s m_z \hat{z}$, and the combined field uses $K_{\text{eff}} = K - \mu_0 M_s^2/2$.

3.5 Derivation of the Zeeman and RKKY Fields

The Zeeman energy $\mathcal{H}_Z = -M_s \int \mathbf{m} \cdot \mathbf{B}_{\text{ext}} dV$ is linear in $\mathbf{m}$:

$$\delta \mathcal{H}_Z = -M_s \int \mathbf{B}_{\text{ext}} \cdot \delta \mathbf{m} dV \quad \implies \quad \mathbf{H}_{\text{ext}} = \mathbf{B}_{\text{ext}} \quad (27)$$

Similarly, the RKKY energy $\mathcal{H}_{\text{RKKY}} = J_{\text{RKKY}} \int \mathbf{m}_{\text{top}} \cdot \mathbf{m}_{\text{bot}} dV$ gives, when varying $\mathbf{m}_{\text{top}}$:

$$\mathbf{H}_{\text{RKKY}}|_{\text{top}} = -\frac{J_{\text{RKKY}}}{M_s} \mathbf{m}_{\text{bot}} = -H_{\text{RKKY}} \mathbf{m}_{\text{bot}} \quad (28)$$

and symmetrically for the bottom layer.

3.6 Physical Interpretation of the LLG Equation

With the effective field defined, the LLG equation

$$\frac{d\mathbf{m}}{dt} = \underbrace{-\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}}}_{\text{precession}} + \underbrace{\alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt}}_{\text{damping}} \quad (29)$$

has a clear energetic interpretation:

- **Precession term** ($-\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}}$): The torque is perpendicular to both $\mathbf{m}$ and $\mathbf{H}_{\text{eff}}$, so the magnetization precesses around the effective field. Since $\mathbf{m} \cdot d\mathbf{m}/dt = 0$ (from $|\mathbf{m}| = 1$) and the torque does no work on the field ($\mathbf{H}_{\text{eff}} \cdot (\mathbf{m} \times \mathbf{H}_{\text{eff}}) = 0$), this term **conserves energy**.
- **Damping term** ($\alpha \mathbf{m} \times d\mathbf{m}/dt$): This produces a torque component that tilts $\mathbf{m}$ toward $\mathbf{H}_{\text{eff}}$. One can show that the energy dissipation rate is:

$$\frac{d\mathcal{H}}{dt} = -\frac{\alpha M_s}{\gamma} \int \left| \frac{d\mathbf{m}}{dt} \right|^2 dV \leq 0 \quad (30)$$

Energy is always dissipated ($d\mathcal{H}/dt \leq 0$), driving the system toward the energy minimum.

Together, the two terms describe a **damped precession**: the magnetization spirals around the effective field while gradually aligning with it. The Hamiltonian defines the energy landscape; the effective field is the local gradient of that landscape; and the LLG equation is the equation of motion that navigates the magnetization toward equilibrium.

The Slonczewski spin-transfer torque terms (Section 6) add a non-conservative driving force from the electric current, which can sustain steady-state motion (e.g. skyrmion translation) against the damping.

4 Effective Field Contributions

The total effective field for each layer is:

$$\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{ex}} + \mathbf{H}_{\text{DMI}} + \mathbf{H}_{\text{anis}} + \mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{RKKY}} + \mathbf{H}_{\text{demag}} \quad (31)$$

All fields are computed in Tesla.

4.1 Exchange Field

On a 2D square lattice with lattice constant a , the discrete exchange field is the lattice Laplacian:

$$\mathbf{H}_{\text{ex}} = \frac{2A_{\text{ex}}}{M_s a^2} (\mathbf{m}_{+x} + \mathbf{m}_{-x} + \mathbf{m}_{+y} + \mathbf{m}_{-y} - 4\mathbf{m}) \quad (32)$$

4.2 Dzyaloshinskii–Moriya Interaction

The discrete interfacial (Néel-type) DMI field:

$$H_{\text{DMI},x} = \frac{D}{M_s a} (m_{+x,z} - m_{-x,z}) \quad (33)$$

$$H_{\text{DMI},y} = \frac{D}{M_s a} (m_{+y,z} - m_{-y,z}) \quad (34)$$

$$H_{\text{DMI},z} = -\frac{D}{M_s a} (m_{+x,x} - m_{-x,x} + m_{+y,y} - m_{-y,y}) \quad (35)$$

where $D > 0$ favors left-handed Néel skyrmions, consistent with the Pt/Co interface.

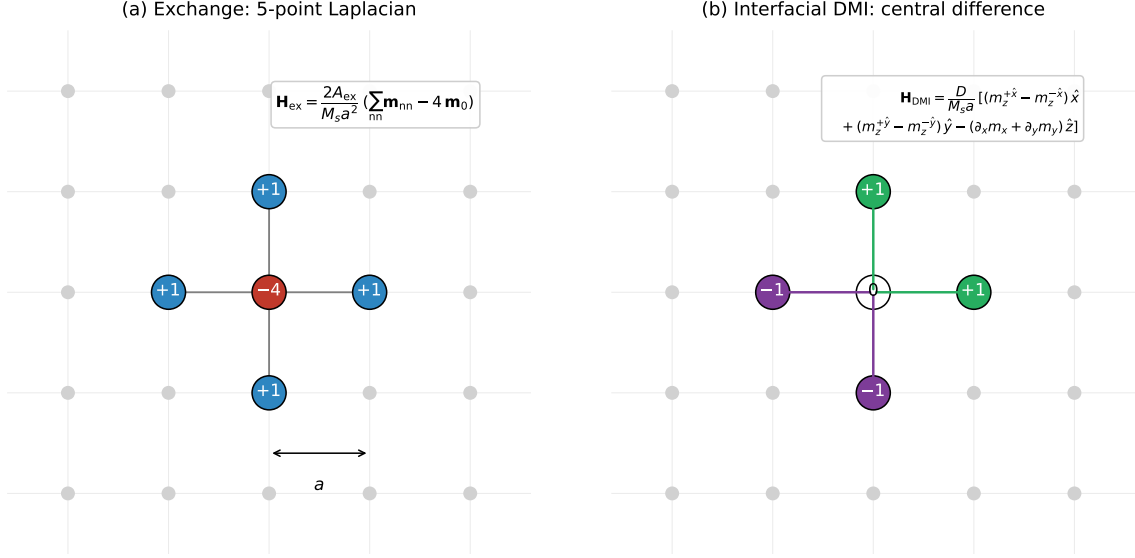


Figure 2: Two-dimensional finite-difference stencils used by the simulator. (a) The exchange field is a 5-point Laplacian on the square lattice: weights $\{+1, +1, +1, +1\}$ on the four nearest neighbours and -4 on the central site, scaled by $2A_{\text{ex}}/(M_s a^2)$. (b) The interfacial DMI field uses a centred first-difference: weights $+1$ on the $+\hat{x}/+\hat{y}$ neighbours, -1 on the $-\hat{x}/-\hat{y}$ neighbours, 0 at the centre. In-plane components couple to gradients of m_z ; the out-of-plane component couples to the in-plane divergence.

4.3 Anisotropy and demagnetization

The perpendicular magnetic anisotropy (PMA) and the demagnetization field share the same functional form $\propto m_z \hat{z}$ in the uniform limit but originate from completely different physics and have completely different locality. Throughout this subsection $\ell \in \{\text{top, bot}\}$ indexes the two Co layers of the SAF, and K_ℓ is the per-layer bare anisotropy $K_\ell = K_{\text{vol}} + 2K_s/t_{\text{Co}}$ of Sec. 2.3 (with distinct values K_{top} and K_{bot} because the Pt and Ru interfaces give different K_s). The codebase supports three levels of approximation, of increasing physical fidelity:

- **Local K_{eff} approximation:** absorbs the uniform- $\mathbf{m}$ part of the demag (i.e. the $-\frac{1}{2}\mu_0 M_s^2$ shape-anisotropy energy density of Sec. 2.4) into a single effective constant $K_{\text{eff},\ell} = K_\ell - \frac{1}{2}\mu_0 M_s^2$ in the PMA term; the demag is not computed explicitly. Valid for textures whose in-plane variation is slow compared with t_{Co} (skyrmion-scale wavelengths satisfy this).
- **Analytic slab demag:** writes the full demag tensor analytically via the thin-film shape factor in Fourier space. Uses the bare K_ℓ in the PMA term and adds the demag separately.
- **Newell demag:** treats every lattice cell as a finite rectangular prism and uses the exact cell-to-cell demag tensor, so it captures the finite-cell corrections that the continuous slab smooths over. Same bare K_ℓ in the PMA term.

The next three subsections give: (i) the discrete PMA formula and the choice of K_ℓ vs $K_{\text{eff},\ell}$ in each branch; (ii) the conceptual distinction between PMA and demagnetization and why they look identical for uniform $\mathbf{m}$; and (iii) the explicit kernel formulas for the three demag approaches.

4.3.1 Perpendicular magnetic anisotropy: discrete form

Uniaxial anisotropy along the film normal ($\hat{z}$). The anisotropy field is local in space (no spatial derivative), so its discrete form is just the continuum formula evaluated at each lattice site (i, j) . In the SAF the top and bottom Co layers are inequivalent (Pt and Ru interfaces); the simulator

carries two distinct anisotropy constants K_{top} and K_{bot} :

$$\mathbf{H}_{\text{anis,top}}(i, j) = \frac{2K_{\text{top}}}{M_s} m_z^{\text{top}}(i, j) \hat{z}, \quad (36)$$

$$\mathbf{H}_{\text{anis,bot}}(i, j) = \frac{2K_{\text{bot}}}{M_s} m_z^{\text{bot}}(i, j) \hat{z}. \quad (37)$$

With the local $N_z = 1$ approximation (Sec. 4.3.3), the shape-anisotropy term is absorbed into an effective constant $K_{\text{eff},\ell} = K_\ell - \frac{1}{2}\mu_0 M_s^2$ ($\ell \in \{\text{top}, \text{bot}\}$), and the discrete combined field becomes:

$$\mathbf{H}_{\text{anis+demag},\ell}(i, j) = \frac{2K_{\text{eff},\ell}}{M_s} m_z^\ell(i, j) \hat{z}. \quad (38)$$

In short: in the slab and Newell branches, the two equations (36)–(37) apply with the bare K_ℓ and the demag is added separately; in the local- $N_z = 1$ branch, equation (38) applies with the combined $K_{\text{eff},\ell}$ and no separate demag is needed. The next subsection explains why this single-constant substitution is exact for spatially uniform $\mathbf{m}$ and where it breaks; the kernel formulas for the slab and Newell branches follow in Sec. 4.3.3.

4.3.2 PMA vs demagnetization: same form, different physics

PMA and the demagnetizing field happen to enter the LLGS right-hand side as terms with the same functional form $\propto m_z \hat{z}$, which makes them visually indistinguishable in the uniform limit and motivates the local $K_{\text{eff},\ell} = K_\ell - \frac{1}{2}\mu_0 M_s^2$ approximation.

Different physical origin. PMA is a *quantum-mechanical interface effect*. Strong spin–orbit coupling at the Co/Pt and Co/Ru contacts hybridizes Co 3d orbitals with the heavy-metal d band; the hybridization lifts the orbital degeneracy in favor of the out-of-plane d_{z^2} -like orbital, and Hund coupling locks the spin to the orbital. The result is an interfacial easy axis along $\hat{z}$, with energy density $-K_s m_z^2/t$ per unit volume for a film of thickness t (Sec. 2.3).

Demagnetization is a *classical magnetostatic effect*. Every magnetic moment generates a dipole field; Maxwell’s equation $\nabla \cdot \mathbf{B} = 0$ forces the magnetization distribution to source surface and volume “charges” that produce a self-field opposing $\mathbf{M}$ (Sec. 2.4). No spin–orbit coupling, no orbital physics — pure classical electromagnetism.

Different locality. The PMA field at lattice site (i, j) depends only on $m_z^\ell(i, j)$ at that same site — it is strictly *local*, as the discrete formula above already shows. The demagnetization field at (i, j) depends on $\mathbf{m}$ at *every* site of *both* layers through a real-space convolution $\mathbf{H}_{\text{demag}} = -\mu_0(\hat{N} * \mathbf{m})$ with a long-range demag tensor $\hat{N}$. Evaluating it requires either an FFT-based convolution (slab kernel, Sec. 4.3.3) or a numerical integration of magnetic surface charges (Newell kernel, Sec. 4.3.3).

Why they look identical in the uniform limit. For spatially uniform $\mathbf{m} = m_z \hat{z}$ (the same configuration analyzed in Sec. 2.4), the demag field of an infinite slab is purely out-of-plane and constant,

$$\mathbf{H}_{\text{demag}}|_{\text{uniform } \mathbf{m}} = -\mu_0 M_s m_z \hat{z}, \quad (39)$$

which is the field equivalent of the $+\frac{1}{2}\mu_0 M_s^2 t_{\text{Co}} \int m_z^2 dS$ energy density of Sec. 2.4. Combined with the bare PMA field $\mathbf{H}_{\text{anis}} = (2K_\ell/M_s) m_z \hat{z}$ (for layer ℓ):

$$\mathbf{H}_{\text{anis}} + \mathbf{H}_{\text{demag}}|_{\text{uniform } \mathbf{m}} = \left(\frac{2K_\ell}{M_s} - \mu_0 M_s \right) m_z \hat{z} = \frac{2K_{\text{eff},\ell}}{M_s} m_z \hat{z}, \quad K_{\text{eff},\ell} = K_\ell - \frac{1}{2}\mu_0 M_s^2. \quad (40)$$

For uniform $\mathbf{m}$ the two effects are mathematically indistinguishable from a single uniaxial anisotropy with constant $K_{\text{eff},\ell}$.

Bare K_ℓ versus effective $K_{\text{eff},\ell}$. Bare K_ℓ is the *physically measured* interfacial PMA constant. It is purely interfacial in origin and has no information about the layer's shape or thickness baked in. For our system:

$$K_{\text{top}} = 1.3106 \times 10^6 \text{ J/m}^3, \quad K_{\text{bot}} = 1.310 \times 10^6 \text{ J/m}^3,$$

both $\sim 10^6 \text{ J/m}^3$, both positive, both *out-of-plane*.

Effective $K_{\text{eff},\ell}$ is the *algebraic combination* $K_\ell - \frac{1}{2}\mu_0 M_s^2$ that emerges from identity (40). It is a numerical artifact of the local-demag approximation, *not* a measurable quantity. For our system,

$$K_{\text{eff},\text{top}} \approx 1.3106 \times 10^6 - 1.285 \times 10^6 \approx 2.6 \times 10^4 \text{ J/m}^3,$$

so a 7% uncertainty in K_ℓ alone exceeds $K_{\text{eff},\ell}$ (Sec. 11.3).

Where the local approximation breaks. To see where the identity (40) stops holding, decompose the in-plane magnetization texture into Fourier modes with wavevector $\mathbf{k}$:

$$\mathbf{m}(\mathbf{r}) = \sum_{\mathbf{k}} \mathbf{m}_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (41)$$

The uniform- $\mathbf{m}$ result of (40) is the $\mathbf{k} = 0$ mode of this expansion. The slab demag kernel component acting on m_z is

$$N_{zz}^{\text{slab}}(\mathbf{k}) = f(|\mathbf{k}| t_{\text{Co}}), \quad f(x) = \frac{1 - e^{-x}}{x}, \quad (42)$$

with the long- and short-wavelength limits

$$\lim_{|\mathbf{k}| t_{\text{Co}} \rightarrow 0} f(|\mathbf{k}| t_{\text{Co}}) = 1, \quad \lim_{|\mathbf{k}| t_{\text{Co}} \rightarrow \infty} f(|\mathbf{k}| t_{\text{Co}}) = 0. \quad (43)$$

The PMA field is local, so it is wavevector-independent; combined with the demag, the per-mode out-of-plane effective field is

$$H_z(\mathbf{k}) = \left(\frac{2K_\ell}{M_s} - \mu_0 M_s f(|\mathbf{k}| t_{\text{Co}}) \right) m_z(\mathbf{k}). \quad (44)$$

At $\mathbf{k} = 0$, $f = 1$ and (44) reduces to $H_z = 2K_{\text{eff},\ell}/M_s m_z(\mathbf{0})$ — the identity (40). At short wavelengths $|\mathbf{k}| t_{\text{Co}} \gg 1$, $f \rightarrow 0$ and the demag piece vanishes:

$$\lim_{|\mathbf{k}| t_{\text{Co}} \rightarrow \infty} H_z(\mathbf{k}) = \frac{2K_\ell}{M_s} m_z(\mathbf{k}) \neq \frac{2K_{\text{eff},\ell}}{M_s} m_z(\mathbf{k}). \quad (45)$$

Substituting $K_{\text{eff},\ell}$ for K_ℓ at those wavelengths is wrong by a factor of $2K_\ell/(2K_{\text{eff},\ell}) \sim 50$ for our parameters ($K_\ell \sim 1.3 \times 10^6$ vs $K_{\text{eff},\ell} \sim 2.6 \times 10^4 \text{ J/m}^3$).

For the present skyrmion problem the relevant length scales are $R \sim 100 \text{ nm}$ and $\Delta_{\text{DW}} \sim 27 \text{ nm}$, so $|\mathbf{k}| t_{\text{Co}} \sim 2\pi t_{\text{Co}}/\Delta \sim 0.30$ at the DW (modestly small) and ~ 0.08 at the skyrmion edge. The slab kernel is close to its long-wavelength limit there ($f \approx 1 - |\mathbf{k}| t_{\text{Co}}/2$), so the local- $K_{\text{eff},\ell}$ approximation produces only a few-percent error. The slab and Newell kernels (Sec. 4.3.3) restore the in-plane components N_{xx} , N_{yy} at $|\mathbf{k}| > 0$, the inter-layer dipolar coupling, and (Newell) the finite-prism corrections at $|\mathbf{k}| t_{\text{Co}} \sim 1$ — effects that the local- $K_{\text{eff},\ell}$ branch discards entirely.

Summary of the three branches.

- **Local- $N_z = 1$ branch:** PMA field uses $K_{\text{eff},\ell}$; no separate demag evaluation. Fast, correct in the long-wavelength uniform- m_z regime, inadequate inside the domain wall where $\mathbf{m}$ is in-plane.
- **Slab or Newell branch:** PMA field uses the bare K_ℓ ; demag is computed separately for every Fourier mode $\mathbf{k}$. No double counting because the bare K_ℓ has no shape correction folded into it.

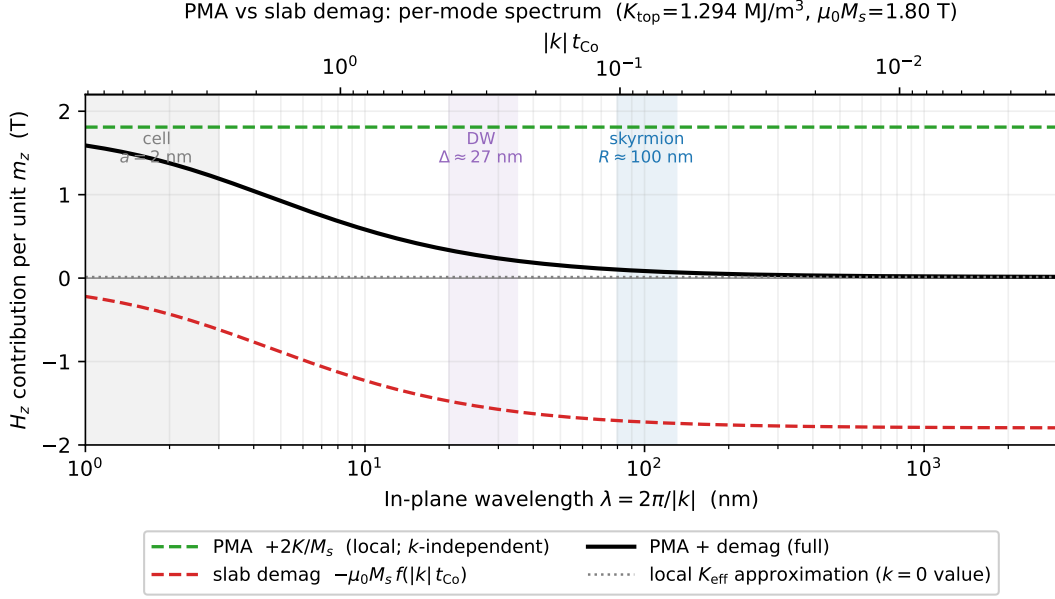


Figure 3: Per-mode comparison of the PMA and slab-demagnetization contributions to the out-of-plane field, as a function of the in-plane wavelength $\lambda = 2\pi/|k|$. Top axis: the dimensionless ratio $|k|t_{\text{Co}}$ for $t_{\text{Co}} = 1.3 \text{ nm}$. The PMA contribution (green, dashed) is constant $\mathbf{H}_{\text{anis}} = 2K/M_s$ because PMA is local in real space. The slab demag contribution (red, dashed) is $\mathbf{H}_{\text{demag}} = -\mu_0 M_s f(|k|t_{\text{Co}})$ with $f(x) = (1 - e^{-x})/x$; it starts at $-\mu_0 M_s$ for uniform $\mathbf{m}$ ($\lambda \rightarrow \infty$) and decays toward zero at the cell scale. Their sum (black, solid) shows that the two nearly cancel at long wavelengths (giving the small $2K_{\text{eff}}/M_s$) but separate sharply at short wavelengths, where the demag has died away and the PMA dominates. The local- K_{eff} approximation (grey dotted) sits at the long-wavelength value $2K_{\text{eff}}/M_s$ for all λ — correct only in the shaded skyrmion regime. Vertical bands mark the relevant length scales: domain-wall width $\Delta_{\text{DW}} \approx 27 \text{ nm}$, skyrmion radius $R \approx 100 \text{ nm}$, and cell scale $a = 2 \text{ nm}$.

4.3.3 Demagnetization kernels

The demagnetizing (stray-dipolar) field is the long-range Coulomb-like field generated by the magnetic “charges” $\rho_m = -\nabla \cdot \mathbf{M}$ and $\sigma_m = \mathbf{M} \cdot \hat{\mathbf{n}}$ at internal and surface discontinuities of $\mathbf{M}$. Three levels of approximation are used in the codebase, in increasing fidelity.

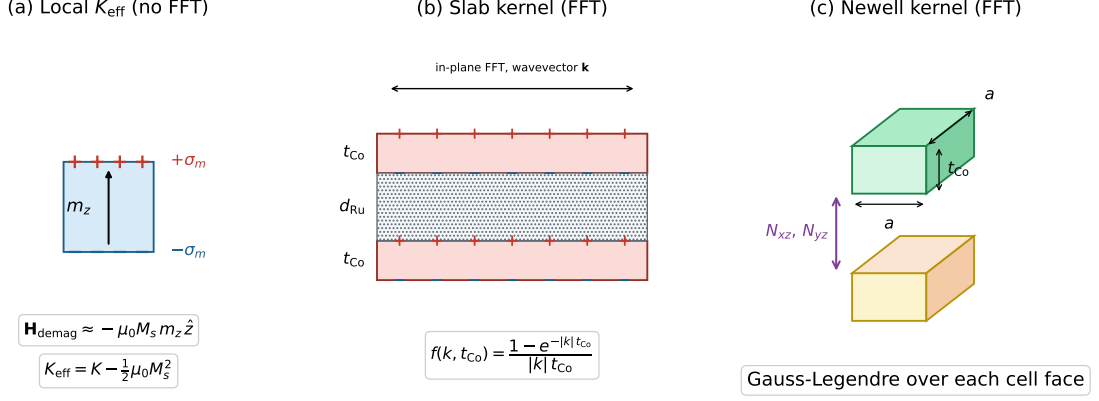


Figure 4: Three demagnetization approaches available in the simulator. (a) Local K_{eff} : each cell sees only the uniform-slab self-demag $-\mu_0 M_s m_z \hat{z}$, which is folded into an effective anisotropy $K_{\text{eff}} = K - \frac{1}{2} \mu_0 M_s^2$. No FFT, per-site, fastest. (b) Slab kernel: each magnetic layer is treated as a continuous slab of thickness t_{Co} with PBC in plane; the demag tensor in $\mathbf{k}$ -space follows the thin-film shape factor $f(k, t_{\text{Co}}) = (1 - e^{-|k| t_{\text{Co}}})/(|k| t_{\text{Co}})$, with self-layer and inter-layer parts. (c) Newell kernel: each lattice cell is a finite $a \times a \times t_{\text{Co}}$ rectangular prism, and the cell-to-cell tensor is the exact field of one uniformly magnetized prism averaged over another.

Level 0: Local $N_z = 1$ (uniform-slab limit). For a uniformly magnetized infinite slab of thickness t_{Co} , the demag field is purely out-of-plane and constant:

$$\mathbf{H}_{\text{demag}} = -\mu_0 M_s m_z \hat{z}. \quad (46)$$

Absorbed into an effective anisotropy:

$$K_{\text{eff}} = K - \frac{\mu_0 M_s^2}{2} \implies \mathbf{H}_{\text{anis+demag}} = \frac{2K_{\text{eff}}}{M_s} m_z \hat{z}. \quad (47)$$

This is the fastest option (no FFT) and is correct in the long-wavelength uniform- m_z regime, but it is *wrong inside the domain wall*, where in-plane $\mathbf{m}$ produces lateral magnetic charges that the local formula ignores.

Level 1: Analytic slab kernel ($\mathbf{k}$ -space). Each layer is treated as a continuous thin slab of thickness t_{Co} with periodic boundary conditions in (x, y) . The demag tensor is analytic in 2D Fourier space. Define the slab shape factor

$$f(|k|, t_{\text{Co}}) = \frac{1 - e^{-|k| t_{\text{Co}}}}{|k| t_{\text{Co}}}, \quad |k| = \sqrt{k_x^2 + k_y^2}, \quad (48)$$

which interpolates between the uniform limit $f \rightarrow 1$ at $|k| t_{\text{Co}} \rightarrow 0$ and the short-wave limit $f \rightarrow 0$ at $|k| t_{\text{Co}} \rightarrow \infty$. Define the structure factor for two parallel slabs of thickness t_{Co} separated edge-to-edge by the non-magnetic spacer d_{Ru} :

$$S(|k|) = \frac{(1 - e^{-|k| t_{\text{Co}}})^2}{2 |k| t_{\text{Co}}} e^{-|k| d_{\text{Ru}}}. \quad (49)$$

The self-layer tensor $\hat{N}_{\text{self}}^{\text{slab}}(\mathbf{k})$ acts on the source layer's own magnetization, the inter-layer tensor $\hat{N}_{\text{inter}}^{\text{slab}}(\mathbf{k})$ couples it to the other layer:

$$N_{xx}^{\text{self}}(\mathbf{k}) = (1 - f(|k|, t_{\text{Co}})) \frac{k_x^2}{|k|^2}, \quad N_{xy}^{\text{self}}(\mathbf{k}) = (1 - f(|k|, t_{\text{Co}})) \frac{k_x k_y}{|k|^2}, \quad (50)$$

$$N_{yy}^{\text{self}}(\mathbf{k}) = (1 - f(|k|, t_{\text{Co}})) \frac{k_y^2}{|k|^2}, \quad N_{zz}^{\text{self}}(\mathbf{k}) = f(|k|, t_{\text{Co}}), \quad (51)$$

$$N_{xx}^{\text{inter}}(\mathbf{k}) = S(|k|) \frac{k_x^2}{|k|^2}, \quad N_{xy}^{\text{inter}}(\mathbf{k}) = S(|k|) \frac{k_x k_y}{|k|^2}, \quad (52)$$

$$N_{yy}^{\text{inter}}(\mathbf{k}) = S(|k|) \frac{k_y^2}{|k|^2}, \quad N_{zz}^{\text{inter}}(\mathbf{k}) = -S(|k|). \quad (53)$$

The two layers are also coupled across components: an in-plane texture in one layer carries volume charges $-\nabla \cdot \mathbf{M}$, whose stray field has an out-of-plane part in the other layer, and vice versa. These cross terms have the same strength S as the other inter-layer components, shifted by a quarter period (the factor i):

$$N_{xz}^{\text{inter}}(\mathbf{k}) = i \frac{k_x}{|k|} S(|k|), \quad N_{yz}^{\text{inter}}(\mathbf{k}) = i \frac{k_y}{|k|} S(|k|). \quad (54)$$

They are odd in the vertical displacement, so they change sign when source and destination layers are swapped. At $\mathbf{k} = 0$ (uniform $\mathbf{m}$): $N_{zz}^{\text{self}} = 1$ and all other components vanish, recovering the local-slab limit. Sign convention:

$$\mathbf{H}_{\text{demag}}(\mathbf{r}) = -\mu_0 M_s (\hat{N}^{\text{self}} * \mathbf{m})(\mathbf{r}) - \mu_0 M_s (\hat{N}^{\text{inter}} * \mathbf{m}_{\text{other}})(\mathbf{r}), \quad (55)$$

where $*$ denotes 2D convolution (multiplication in $\mathbf{k}$). Cost per call: 6 forward + 6 inverse FFTs.

Level 2: Newell numerical kernel (finite prisms). Each lattice cell is a uniformly magnetized rectangular prism of size $a \times a \times t_{\text{Co}}$. The cell-to-cell demag tensor is the field that one prism's surface charges produce, averaged over the volume of another prism. For source magnetization $\mathbf{M} = M_u \hat{u}$ ($u \in \{x, y, z\}$), surface charges $\pm M_u$ live on the two u -faces. The volume-averaged field at a destination cell with center-to-center displacement $\mathbf{R}$ is

$$\langle H_i^{\text{Newell}}(\mathbf{R}) \rangle_{V_{\text{dest}}} = \frac{M_u}{V_{\text{dest}}} \int_{V_{\text{dest}}} d^3r \left[\int_{+u \text{ face}} \frac{(\mathbf{r} - \mathbf{r}')_i}{4\pi |\mathbf{r} - \mathbf{r}'|^3} dA' - \int_{-u \text{ face}} \frac{(\mathbf{r} - \mathbf{r}')_i}{4\pi |\mathbf{r} - \mathbf{r}'|^3} dA' \right]. \quad (56)$$

This 5-D integral (2-D source surface $\times$ 3-D destination volume) has an exact closed form, due to Newell, Williams and Dunlop [5]: two auxiliary functions f and g (for diagonal and off-diagonal components) are combined in a second-difference over the 27 corner offsets of the two prisms. The closed form is exact but subtracts nearly equal numbers, so it loses precision far from the source. Beyond about 40 cells the prism is indistinguishable from a point dipole, and the dipole tensor is used instead. The same integral can also be evaluated by adaptive Gauss-Legendre quadrature (the MuMax3 recipe [4]), which serves as an independent cross-check.

With periodic boundaries, every cell also feels the periodic copies of the box. The tensor therefore adds the contributions of the nearest image boxes, which sit at least one box length away and are well described by the dipole form.

For the self-cell $\mathbf{R} = \mathbf{0}$ the source and destination coincide. Its demag factors are given directly by Aharoni's closed form for a rectangular prism [6]:

$$N_c^{\text{Aharoni}}(a, b, c) = \frac{1}{\pi} \left[\frac{b^2 - c^2}{2bc} \log \frac{R_{abc} - a}{R_{abc} + a} + \frac{a^2 - c^2}{2ac} \log \frac{R_{abc} - b}{R_{abc} + b} + \frac{b}{2c} \log \frac{R_{ab} + a}{R_{ab} - a} + \dots \right], \quad (57)$$

where $R_{abc} = \frac{1}{2}\sqrt{a^2 + b^2 + c^2}$, $R_{ab} = \frac{1}{2}\sqrt{a^2 + b^2}$, and the remaining terms close the formula (full expression: Aharoni [6], Eq. 1). This evaluates $N_{xx,yy,zz}^{\text{self}}(\mathbf{R} = 0)$ exactly; off-diagonals vanish by the cell mirror symmetries.

As in the slab kernel, the inter-layer tensor includes the z - xy cross-couplings:

$$N_{xz}^{\text{inter}}, N_{yz}^{\text{inter}} \neq 0 \quad (\text{odd in the inter-layer } z\text{-displacement}). \quad (58)$$

These add the contributions $-\mu_0 M_s (N_{xz}^{\text{inter}} m_z^{\text{other}})$ to H_x^{dest} (and symmetric variants); the sign flips when swapping source and destination layers because $N_{xz}(\mathbf{R}, +Z) = -N_{xz}(\mathbf{R}, -Z)$.

After real-space construction over all (n_x, n_y) lattice displacements, each tensor component is FFT-transformed to $\mathbf{k}$ -space so the per-step demag evaluation remains 6 forward + 6 inverse FFTs — the same cost as the slab kernel. The result is a pair of demag tensors $\hat{N}^{\text{Newell,self}}(\mathbf{k})$ and $\hat{N}^{\text{Newell,inter}}(\mathbf{k})$ that play the same role as their slab counterparts but are obtained *numerically* (no closed-form $f(|\mathbf{k}| t_{\text{Co}})$ shape factor). Schematically,

$$N_{xx}^{\text{Newell,self}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{xx}^{\text{self}}(\mathbf{R})\}, \quad N_{xy}^{\text{Newell,self}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{xy}^{\text{self}}(\mathbf{R})\}, \quad (59)$$

$$N_{yy}^{\text{Newell,self}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{yy}^{\text{self}}(\mathbf{R})\}, \quad N_{zz}^{\text{Newell,self}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{zz}^{\text{self}}(\mathbf{R})\}, \quad (60)$$

$$N_{xx}^{\text{Newell,inter}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{xx}^{\text{inter}}(\mathbf{R})\}, \quad N_{xy}^{\text{Newell,inter}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{xy}^{\text{inter}}(\mathbf{R})\}, \quad (61)$$

$$N_{yy}^{\text{Newell,inter}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{yy}^{\text{inter}}(\mathbf{R})\}, \quad N_{zz}^{\text{Newell,inter}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{zz}^{\text{inter}}(\mathbf{R})\}, \quad (62)$$

$$N_{xz}^{\text{Newell,inter}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{xz}^{\text{inter}}(\mathbf{R})\}, \quad N_{yz}^{\text{Newell,inter}}(\mathbf{k}) = \mathcal{F}\{\bar{N}_{yz}^{\text{inter}}(\mathbf{R})\}, \quad (63)$$

where $\bar{N}_{ij}^{\text{layer}}(\mathbf{R})$ is the real-space cell-cell tensor from Eq. (56) for $\mathbf{R} \neq \mathbf{0}$ and the Aharoni override at $\mathbf{R} = \mathbf{0}$, and $\mathcal{F}$ is the 2-D discrete Fourier transform. The structure is the same as the slab kernel, including the inter-layer cross-terms N_{xz}^{inter} and N_{yz}^{inter} . The sign convention is also the same:

$$\mathbf{H}_{\text{demag}}(\mathbf{r}) = -\mu_0 M_s (\hat{N}^{\text{Newell,self}} * \mathbf{m})(\mathbf{r}) - \mu_0 M_s (\hat{N}^{\text{Newell,inter}} * \mathbf{m}_{\text{other}})(\mathbf{r}). \quad (64)$$

Choosing a level. No level is universally best; the choice depends on the question. At skyrmion-scale wavelengths $|\mathbf{k}| t_{\text{Co}} \sim 2\pi t_{\text{Co}}/R \sim 0.08 \ll 1$ the slab and Newell formulations nearly coincide, and the local level captures the dominant shape anisotropy. The Newell kernel matters at shorter wavelengths, where the finite-cell corrections (in-plane components at $|\mathbf{k}| > 0$ and prism-shape effects at $|\mathbf{k}| t_{\text{Co}} \sim 1$) become physically relevant.

Finite samples. Fourier-space multiplication is a *circular* convolution: it describes a periodic sample. An isolated sample is modelled by zero-padding the grid to twice its size, so the convolution no longer wraps around. Padding both in-plane directions describes a finite magnet such as a disc or a rectangle; padding only across the width describes a racetrack, periodic along its length and free at its two edges.

4.4 Zeeman Field

A uniform external magnetic field $\mathbf{H}_{\text{ext}}$ (in Tesla), identical at all lattice sites.

4.5 RKKY Interlayer Coupling

The antiferromagnetic RKKY interaction promotes antiparallel alignment between the two Co layers across the Ru/Pt spacer. Like the PMA, it is *local* in the in-plane lattice: the field on the top spin at site (i, j) depends only on the bottom spin at the same in-plane position, and vice versa. The discrete form is the continuum formula evaluated site-by-site:

$$\mathbf{H}_{\text{RKKY,top}}(i, j) = -H_{\text{RKKY}} \mathbf{m}^{\text{bot}}(i, j), \quad (65)$$

$$\mathbf{H}_{\text{RKKY,bot}}(i, j) = -H_{\text{RKKY}} \mathbf{m}^{\text{top}}(i, j), \quad (66)$$

with $H_{\text{RKKY}} > 0$ (antiferromagnetic) carried as a single scalar in Tesla.

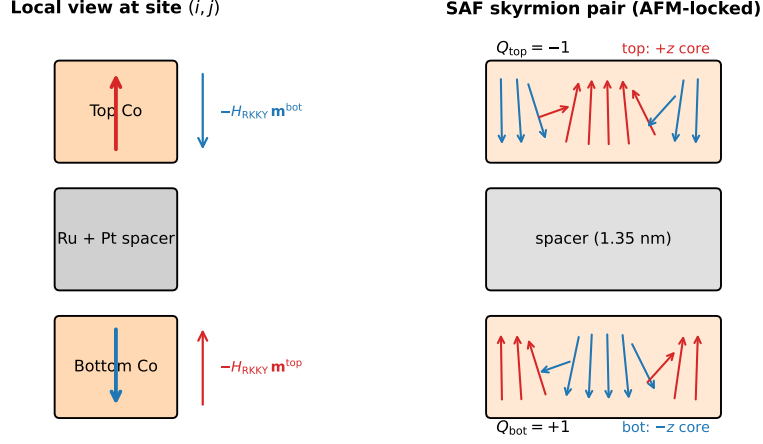


Figure 5: Antiferromagnetic RKKY interlayer coupling in the SAF stack. The Ru/Pt spacer mediates an indirect exchange interaction between the two Co layers whose oscillating sign is tuned to favour antiparallel alignment at the chosen thickness $d_{\text{Ru}} + d_{\text{Pt}} = 1.35$ nm. Each top spin $\mathbf{m}^{\text{top}}(i, j)$ feels an effective field $-H_{\text{RKKY}} \mathbf{m}^{\text{bot}}(i, j)$ pointing opposite to the bottom spin at the same in-plane site, and vice versa. The result is the AFM-locked skyrmion pair shown on the right: identical lateral position, opposite core polarities, opposite topological charges. The Magnus and skyrmion-Hall contributions cancel between layers, allowing the SAF skyrmion to move along the current direction with no transverse drift.

5 Landau–Lifshitz–Gilbert–Slonczewski Equation (LLGS)

The magnetization dynamics of each layer are governed by the LLGS equation. For the unit magnetization vector $\mathbf{m} = \mathbf{M}/M_s$:

$$\frac{d\mathbf{m}}{dt} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} + \gamma H_{\text{DL}} \mathbf{m} \times (\hat{\mathbf{p}} \times \mathbf{m}) + \gamma H_{\text{FL}} \hat{\mathbf{p}} \times \mathbf{m} \quad (67)$$

where:

- γ is the gyromagnetic ratio (rad/s/T),
- α is the Gilbert damping constant,
- $\mathbf{H}_{\text{eff}}$ is the effective magnetic field (Tesla),
- $H_{\text{DL}} = \chi_{\text{DL}} \cdot J(t)$ is the damping-like (DL) SOT field,
- $H_{\text{FL}} = \chi_{\text{FL}} \cdot J(t)$ is the field-like (FL) SOT field,
- $\hat{\mathbf{p}}$ is the spin polarization direction ($\hat{\mathbf{p}} = \hat{\mathbf{z}} \times \hat{\mathbf{J}}$),
- $\mathbf{J} = J(t) \hat{\mathbf{J}}$ is the current density: $J(t)$ is the magnitude (A/m²), given as a current pulse in time, and $\hat{\mathbf{J}}$ is the unit vector along the current flow,
- $\hat{\mathbf{z}}$ is the film-normal unit vector (out-of-plane).

These are the standard Slonczewski torques. Their magnitude is proportional to $|\hat{\mathbf{p}} \times \mathbf{m}| = \sin \vartheta$, where ϑ is the angle between $\mathbf{m}$ and $\hat{\mathbf{p}}$, so they are largest when $\mathbf{m} \perp \hat{\mathbf{p}}$ and vanish when $\mathbf{m}$ is parallel to the spin polarization.

5.1 Explicit Form

The implicit form (67) contains $d\mathbf{m}/dt$ on both sides through the Gilbert damping term. To convert it to a fully explicit ODE solvable by Runge–Kutta, we eliminate the right-hand-side $d\mathbf{m}/dt$ algebraically. The standard procedure follows Mallinson [2] for the LLG part and Khvalkovskiy *et al.* [3] for the SOT extension.

Step 1: Compact form. Define $\mathbf{s} = \hat{\mathbf{p}} \times \mathbf{m}$ and group the right-hand side of (67) as

$$\frac{d\mathbf{m}}{dt} = \mathbf{F} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt}, \quad \mathbf{F} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \gamma H_{\text{DL}} (\mathbf{m} \times \mathbf{s}) + \gamma H_{\text{FL}} \mathbf{s}. \quad (68)$$

Step 2: Cross with $\mathbf{m}$. Take $\mathbf{m} \times$ of both sides of (68):

$$\mathbf{m} \times \frac{d\mathbf{m}}{dt} = \mathbf{m} \times \mathbf{F} + \alpha \mathbf{m} \times \left(\mathbf{m} \times \frac{d\mathbf{m}}{dt} \right). \quad (69)$$

The triple product is simplified by the BAC-CAB identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$. Using $|\mathbf{m}| = 1$ (so $\mathbf{m} \cdot \mathbf{m} = 1$) and $\mathbf{m} \cdot d\mathbf{m}/dt = \frac{1}{2}d|\mathbf{m}|^2/dt = 0$:

$$\mathbf{m} \times \left(\mathbf{m} \times \frac{d\mathbf{m}}{dt} \right) = \mathbf{m} \left(\mathbf{m} \cdot \frac{d\mathbf{m}}{dt} \right) - \frac{d\mathbf{m}}{dt} (\mathbf{m} \cdot \mathbf{m}) = -\frac{d\mathbf{m}}{dt}. \quad (70)$$

Substituting (70) into (69):

$$\mathbf{m} \times \frac{d\mathbf{m}}{dt} = \mathbf{m} \times \mathbf{F} - \alpha \frac{d\mathbf{m}}{dt}. \quad (71)$$

Step 3: Solve for $d\mathbf{m}/dt$. Insert (71) back into the $\alpha \mathbf{m} \times d\mathbf{m}/dt$ term of (68):

$$\frac{d\mathbf{m}}{dt} = \mathbf{F} + \alpha \left(\mathbf{m} \times \mathbf{F} - \alpha \frac{d\mathbf{m}}{dt} \right), \quad (72)$$

collect the $d\mathbf{m}/dt$ terms on the left, and divide by $1 + \alpha^2$:

$$\boxed{\frac{d\mathbf{m}}{dt} = \frac{1}{1 + \alpha^2} \left[\mathbf{F} + \alpha \mathbf{m} \times \mathbf{F} \right]}. \quad (73)$$

Step 4: Expand $\mathbf{m} \times \mathbf{F}$. The three contributions to $\mathbf{F}$ in (68) give:

$$\mathbf{m} \times (-\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}}) = -\gamma \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}), \quad (74)$$

$$\mathbf{m} \times (\gamma H_{\text{DL}} \mathbf{m} \times \mathbf{s}) = \gamma H_{\text{DL}} \mathbf{m} \times (\mathbf{m} \times \mathbf{s}) = -\gamma H_{\text{DL}} \mathbf{s}, \quad (75)$$

$$\mathbf{m} \times (\gamma H_{\text{FL}} \mathbf{s}) = \gamma H_{\text{FL}} \mathbf{m} \times \mathbf{s}, \quad (76)$$

where (75) again uses BAC-CAB with $\mathbf{m} \cdot \mathbf{s} = 0$ (since $\mathbf{s} \propto \hat{\mathbf{p}} \times \mathbf{m}$ is perpendicular to $\mathbf{m}$): $\mathbf{m} \times (\mathbf{m} \times \mathbf{s}) = \mathbf{m}(\mathbf{m} \cdot \mathbf{s}) - \mathbf{s} = -\mathbf{s}$.

Step 5: Collect coefficients. Assembling $\mathbf{F} + \alpha \mathbf{m} \times \mathbf{F}$ and dividing by $1 + \alpha^2$:

$$\boxed{\frac{d\mathbf{m}}{dt} = \frac{\gamma}{1 + \alpha^2} \left[-\mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) + (H_{\text{DL}} + \alpha H_{\text{FL}}) (\mathbf{m} \times \mathbf{s}) + (H_{\text{FL}} - \alpha H_{\text{DL}}) \mathbf{s} \right]}. \quad (77)$$

The coefficients $H_{\text{DL}} + \alpha H_{\text{FL}}$ and $H_{\text{FL}} - \alpha H_{\text{DL}}$ are the standard “mixed” SOT coefficients of the explicit form [3]; the cross-coupling between DL and FL through α arises entirely from Step 3’s $\alpha \mathbf{m} \times \mathbf{F}$ term and vanishes at $\alpha = 0$. The implementation in `src/skyrmion_simulator/simulator/integrator.llgs_rhs` precomputes $\gamma' = \gamma/(1 + \alpha^2)$, $c_{\text{DL}} = H_{\text{DL}} + \alpha H_{\text{FL}}$, $c_{\text{FL}} = H_{\text{FL}} - \alpha H_{\text{DL}}$, and evaluates (77) directly without further algebra.

5.2 Thermal Fluctuations

At finite temperature the magnetization is buffeted by the thermal bath. Following Brown, this is modelled by a random thermal field $\mathbf{h}(t)$ added to $\mathbf{H}_{\text{eff}}$ in the LLGS equation, so it enters both the precession and the damping terms. Each Cartesian component in each cell is Gaussian white noise,

$$\langle h_i(t) \rangle = 0, \quad \langle h_i(t) h_j(t') \rangle = \sigma^2 \delta_{ij} \delta(t - t'), \quad \sigma^2 = \frac{2\alpha k_B T}{\gamma M_s V_{\text{cell}}}. \quad (78)$$

The strength σ^2 follows from the fluctuation–dissipation theorem: it is the value for which the dynamics relax to the Boltzmann distribution at temperature T . Larger damping means stronger coupling to the bath, and a smaller cell volume means larger relative fluctuations. Over a time step Δt the field is constant with standard deviation $\sigma/\sqrt{\Delta t}$, so its time integral has variance $\sigma^2 \Delta t$, as white noise requires.

The noise multiplies $\mathbf{m}$ inside the cross products, so the stochastic equation must be read in the Stratonovich sense, the interpretation that preserves $|\mathbf{m}| = 1$. The Heun predictor–corrector scheme converges to the Stratonovich solution: it averages the slopes at the start of the step and at a forward-Euler prediction, both evaluated with the *same* noise sample.

A current also heats the film. To lowest order the temperature rises quadratically with the current density, $T = T_{\text{sub}} + R_{\text{th}} J^2$, where T_{sub} is the substrate temperature and R_{th} a thermal resistance.

6 Spin-Orbit Torques

The spin-orbit torques (SOT) are the mechanism by which an electric current drives skyrmion motion. They originate from the **spin Hall effect** (SHE) in the heavy-metal Pt layers adjacent to the ferromagnetic Co.

6.1 Spin Hall Effect

When an in-plane charge current density $\mathbf{J}$ flows through the Pt layer, the strong spin-orbit coupling deflects spin-up and spin-down electrons in opposite transverse directions. This generates a transverse *spin current* that accumulates at the Pt/Co interface. The spin polarization direction is perpendicular to both the current and the film normal:

$$\hat{\mathbf{p}} = \hat{\mathbf{z}} \times \hat{\mathbf{J}} \quad (79)$$

For current along $\hat{x}$, the spin polarization is $\hat{\mathbf{p}} = \hat{y}$. The efficiency of the conversion from charge current to spin current is characterized by the spin Hall angle θ_{SH} , which is encapsulated in the experimentally measured SOT coefficients χ_{DL} and χ_{FL} .

6.2 SOT Torques in the LLGS Equation

The spin current accumulated at the interface exerts two distinct torques on the Co magnetization, which enter as the Slonczewski terms in the LLGS equation:

- **Damping-like (DL) torque:** $\tau_{\text{DL}} = H_{\text{DL}} \mathbf{m} \times (\hat{\mathbf{p}} \times \mathbf{m})$. This acts like an additional damping (or anti-damping) that pushes the magnetization toward or away from $\hat{\mathbf{p}}$. It is the dominant torque driving skyrmion motion.
- **Field-like (FL) torque:** $\tau_{\text{FL}} = H_{\text{FL}} (\hat{\mathbf{p}} \times \mathbf{m})$. This is equivalent to an effective magnetic field along $\hat{\mathbf{p}}$ and is typically smaller in magnitude.

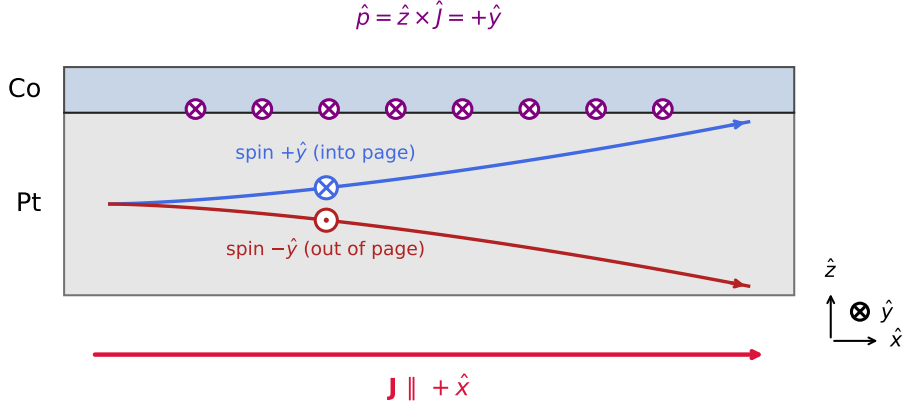


Figure 6: Spin Hall effect in the Pt/Co bilayer (side view in the x - z plane; right-handed axes with $\hat{y}$ into the page). An in-plane charge current $\mathbf{J} \parallel +\hat{x}$ in the heavy metal Pt deflects spin- $+\hat{y}$ electrons toward $+\hat{z}$ and spin- $-\hat{y}$ electrons toward $-\hat{z}$. The transverse spin current accumulates spin polarization $\hat{\mathbf{p}} = \hat{\mathbf{z}} \times \hat{\mathbf{J}} = +\hat{y}$ at the Pt/Co interface (row of $\otimes$ markers), which drives the SOT torques on the Co magnetization.

Both torques scale with $\sin \vartheta$, the sine of the angle between $\mathbf{m}$ and $\hat{\mathbf{p}}$. The SOT effective field magnitudes are proportional to the applied current density, which follows the current pulse $J(t)$:

$$H_{\text{DL}} = \chi_{\text{DL}} \cdot J(t), \quad H_{\text{FL}} = \chi_{\text{FL}} \cdot J(t) \quad (80)$$

where χ_{DL} and χ_{FL} are the SOT coefficients (in $\text{T A}^{-1} \text{m}^2$) measured experimentally. These coefficients encapsulate the spin Hall angle, the spin diffusion length, and the interface spin transparency into a single effective parameter.

6.3 SOT Terms in the Explicit LLGS

In the explicit form of the LLGS equation (77), the SOT terms appear as the last two contributions:

$$\frac{d\mathbf{m}}{dt} = \frac{\gamma}{1 + \alpha^2} \left[\underbrace{-\mathbf{m} \times \mathbf{H} - \alpha \mathbf{m} \times (\mathbf{m} \times \mathbf{H})}_{\text{precession + damping}} + \underbrace{(H_{\text{DL}} + \alpha H_{\text{FL}})(\mathbf{m} \times \mathbf{s}) + (H_{\text{FL}} - \alpha H_{\text{DL}})\mathbf{s}}_{\text{SHE-driven SOT}} \right] \quad (81)$$

where $\mathbf{s} = \hat{\mathbf{p}} \times \mathbf{m}$. Without the SHE ($J = 0$), the SOT terms vanish and the skyrmion relaxes to its equilibrium configuration. With nonzero current, the DL-SOT exerts a net force on the skyrmion domain wall, driving translational motion. In the SAF geometry, the opposite topological charges of the two layers cancel the Magnus force, resulting in motion along the current direction with no skyrmion Hall effect [1].

6.4 Topological Spin Hall Torque

The SOT discussed above is an *interface* effect generated in the heavy-metal Pt layer. A second current-driven torque exists in the *bulk* of the Co layer, originating from the Berry phase that conduction electrons pick up as they adiabatically follow a non-trivial spin texture [8, 9]. This *topological spin Hall* (TSH) torque is the Pham 2024 mechanism and switches on whenever the texture carries a non-zero local skyrmion-number density.

Topological charge density. Define the local skyrmion-number density (Pontryagin density):

$$N_{xy}(\mathbf{r}, t) = \mathbf{m} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m}). \quad (82)$$

Units of $1/\text{m}^2$. Integrated over the layer, $(4\pi)^{-1} \int N_{xy} dA$ gives the total topological charge Q (Section 8). N_{xy} is concentrated in the domain-wall ring of the skyrmion and vanishes in the uniformly magnetized background.

Berry-phase spin current. An electron with spin polarization P drifting through a region of nonzero N_{xy} acquires a Berry phase that deflects its trajectory transversely to the applied charge current. The resulting topological Hall current is itself spin-polarized; it re-deposits angular momentum onto the underlying magnetization through a spin-transfer-torque-like coupling. The leading-order contribution to $d\mathbf{m}/dt$ is [9]:

$$\left. \frac{d\mathbf{m}}{dt} \right|_{\text{TSH}} = -b_j \lambda^2 N_{xy}(\mathbf{r}, t) \partial_y \mathbf{m} \quad (83)$$

with the spin-drift velocity

$$b_j = \frac{\mu_B}{q_e} \frac{P J(t)}{M_s} \quad [\text{m/s}], \quad (84)$$

where μ_B is the Bohr magneton, q_e the elementary charge, P the conduction-electron spin polarization, $J(t)$ the applied charge current density, and M_s the saturation magnetization. λ^2 is the topological coupling length squared (m^2); it sets the effective spin-diffusion area relevant to the topological Hall current and is treated as a material parameter.

Dimensions. The product $b_j[\text{m/s}] \cdot \lambda^2[\text{m}^2] \cdot N_{xy}[1/\text{m}^2] \cdot \partial_y \mathbf{m}[1/\text{m}]$ evaluates to $1/\text{s}$, the correct unit for $d\mathbf{m}/dt$.

Direction. If the charge current is taken along $\hat{x}$, the topological Hall current flows along $\hat{y}$, so the resulting spin transfer couples to the magnetization gradient *along* $\hat{y}$ — $\partial_y \mathbf{m}$. Rotating the current direction would permute this through $\hat{x} \leftrightarrow \hat{y}$ in (83).

Where the torque acts. In the uniformly magnetized background $N_{xy} = 0$ and the TSH torque vanishes identically. Inside the skyrmion ring, $|N_{xy}|$ peaks at the domain wall, so the TSH torque is localized there. The DL-SOT in Eq. (77) drives the entire skyrmion; the TSH adds a wall-localized push along the direction set by $\partial_y \mathbf{m}$. The two torques are **independent and additive** — TSH does not replace SOT.

Sign convention and net force. With $J > 0$ along $+\hat{x}$, $P > 0$, and a skyrmion of charge $Q = -1$ in the top layer (the convention used in the codebase), N_{xy} is predominantly negative in the wall. Equation (83) then produces a transverse push consistent with the topological Hall sign (electrons deflect toward $+\hat{y}$, magnetization responds in $-\hat{y}$ via spin angular-momentum conservation). In the SAF the opposite skyrmion charges on the two layers send the TSH push in opposite $\hat{y}$ directions; the net translational contribution along $\hat{x}$ survives, but the transverse components cancel just as the Magnus force does.

Discrete-lattice implementation. On the simulation grid, $\partial_{x,y} \mathbf{m}$ are evaluated as second-order central differences. N_{xy} is the elementwise scalar $\mathbf{m} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m})$. The TSH contribution is added to $d\mathbf{m}/dt$ after the LLGS right-hand side:

$$d\mathbf{m}/dt += -b_j \lambda^2 N_{xy} \partial_y \mathbf{m}.$$

When the mechanism is disabled ($\lambda^2 = 0$), the SOT-only baseline is recovered.

7 Magnus Force and the Thiele Equation

The Magnus force is not a separate term in the LLGS equation — it *emerges* from it when skyrmion motion is analyzed as a collective coordinate problem. Assume the skyrmion is a rigid texture translated by its centre-of-mass coordinate $\mathbf{R}(t)$:

$$\mathbf{m}(\mathbf{r}, t) = \mathbf{m}_0(\mathbf{r} - \mathbf{R}(t)), \quad \partial_t \mathbf{m} = -(\mathbf{v} \cdot \nabla) \mathbf{m}, \quad \mathbf{v} \equiv \dot{\mathbf{R}}. \quad (85)$$

Substituting (85) into the explicit LLGS (77) gives the rigid-translation equation of motion,

$$-(\mathbf{v} \cdot \nabla) \mathbf{m} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times [-(\mathbf{v} \cdot \nabla) \mathbf{m}] - \gamma \boldsymbol{\tau}_{\text{SOT}}. \quad (86)$$

Take the cross product with $\partial_i \mathbf{m}$ (with $i \in \{x, y\}$) and integrate over the film plane. Three standard identities collapse the result:

$$\int \mathbf{m} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m}) dA = 4\pi Q, \quad (87)$$

$$G_k \equiv \frac{1}{2} \varepsilon_{kij} \int \mathbf{m} \cdot (\partial_i \mathbf{m} \times \partial_j \mathbf{m}) dA = 4\pi Q \delta_{kz}, \quad (88)$$

$$D_{ij} \equiv \int \partial_i \mathbf{m} \cdot \partial_j \mathbf{m} dA, \quad (89)$$

defining the gyrovector $\mathbf{G} = 4\pi Q \hat{z}$ (the factor $\frac{1}{2}$ compensates the two orderings $ij = xy, yx$ in the sum) and the dissipation tensor $[\mathbf{D}]$. The driving force from the SOT is

$$F_i^{\text{SOT}} \equiv -\gamma \int \boldsymbol{\tau}_{\text{SOT}} \cdot \partial_i \mathbf{m} dA. \quad (90)$$

Collecting the gyrotopic, dissipative, and driving terms gives the **Thiele equation**:

$$\mathbf{F}_{\text{SOT}} + \mathbf{G} \times \mathbf{v} + \alpha [\mathbf{D}] \mathbf{v} = 0. \quad (91)$$

The three terms represent:

- $\mathbf{F}_{\text{SOT}}$: the driving force from the spin-orbit torque, directed along the current.
- $\mathbf{G} \times \mathbf{v}$: the **Magnus force**, directed perpendicular to the skyrmion velocity. The gyrovector is $\mathbf{G} = G \hat{z}$ with $G = 4\pi Q$, where Q is the topological charge. This force is analogous to the Magnus effect on a spinning ball and is a direct consequence of the skyrmion's non-trivial topology.
- $\alpha [\mathbf{D}] \mathbf{v}$: the dissipative drag from Gilbert damping, where $[\mathbf{D}]$ is the dissipation tensor.

The Magnus force deflects the skyrmion sideways relative to the driving force — this is the **skyrmion Hall effect**. From (91), the skyrmion Hall angle is:

$$\tan \theta_{\text{SkHE}} = \frac{v_y}{v_x} = \frac{G}{\alpha D} \quad (92)$$

where D is the diagonal component of the dissipation tensor.

In a **single ferromagnetic layer** with $Q = \pm 1$, the gyrovector $G = \pm 4\pi \neq 0$, and the skyrmion is deflected at a significant angle to the current (experimentally up to 60° – 85° at high velocities [1]).

In the **SAF**, the two layers carry opposite topological charges ($Q_{\text{top}} = -Q_{\text{bot}}$), so their gyrovector cancel:

$$G_{\text{net}} = G_{\text{top}} + G_{\text{bot}} = 4\pi Q_{\text{top}} + 4\pi Q_{\text{bot}} = 0 \quad (93)$$

The Magnus forces on the two layers point in opposite directions and cancel exactly. The skyrmion moves straight along the current direction with $\theta_{\text{SkHE}} = 0$. This is the key advantage of the SAF geometry for skyrmion-based devices, as confirmed experimentally ($\theta_{\text{SkHE}} = 6.4^\circ \pm 16.4^\circ$ [1]).

Closed-form speeds. For a compensated SAF the Magnus term drops out, and the Thiele equation reduces to a balance between the driving force and the dissipative drag, $\mathbf{F} = \alpha[\mathbf{D}] \mathbf{v}$. Evaluating both for a rigid skyrmion of radius R whose wall has the 1-D profile of Appendix A, with width Δ , gives the damping-like SOT speed [1]

$$v_{\text{SOT}} = \frac{\pi \gamma H_{\text{DL}} R}{2\alpha (R/\Delta + \Delta/R)}. \quad (94)$$

The denominator comes from the dissipation tensor. For a large skyrmion ($R \gg \Delta$) the speed saturates at $\pi\gamma H_{\text{DL}}\Delta/(2\alpha)$: both the force and the drag grow with the wall length, so their ratio becomes independent of R . The topological spin Hall torque of Section 6.4 adds, with the same drag,

$$v_{\text{TSH}} = -\frac{b_j \lambda^2}{\alpha (R/\Delta + \Delta/R)} \int N_{xy}^2 dA, \quad (95)$$

which is linear in λ^2 . The integral of N_{xy}^2 decreases as the skyrmion grows, because the topological density spreads over a longer wall, so v_{TSH} is largest for small skyrmions and is a correction to v_{SOT} .

8 Topological Charge

The topological charge (skyrmion number) is:

$$Q = \frac{1}{4\pi} \int \mathbf{m} \cdot \left(\frac{\partial \mathbf{m}}{\partial x} \times \frac{\partial \mathbf{m}}{\partial y} \right) dx dy \quad (96)$$

For a single skyrmion, $Q = \pm 1$. In the SAF, the two layers carry opposite topological charges ($Q_{\text{top}} = -Q_{\text{bot}}$), leading to a net topological charge of zero and hence no skyrmion Hall effect.

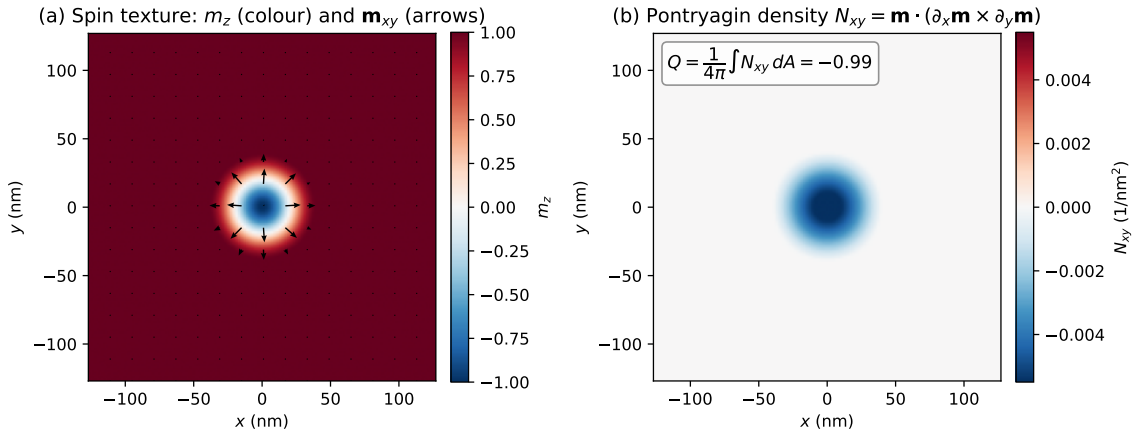


Figure 7: Topological charge of an isolated Néel skyrmion ($Q = -1$ by our analytic convention with $\theta(0) = \pi$). (a) Real-space texture: m_z in colour (blue core, $m_z = -1$; red background, $m_z = +1$), in-plane component $\mathbf{m}_{xy}$ as black arrows on a subsampled grid (Néel-outward chirality). (b) Pontryagin density $N_{xy} = \mathbf{m} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m})$; its Riemann sum recovers $Q \simeq -1$ (annotated). In the SAF the two layers carry opposite-sign N_{xy} , so the layer-summed charge vanishes.

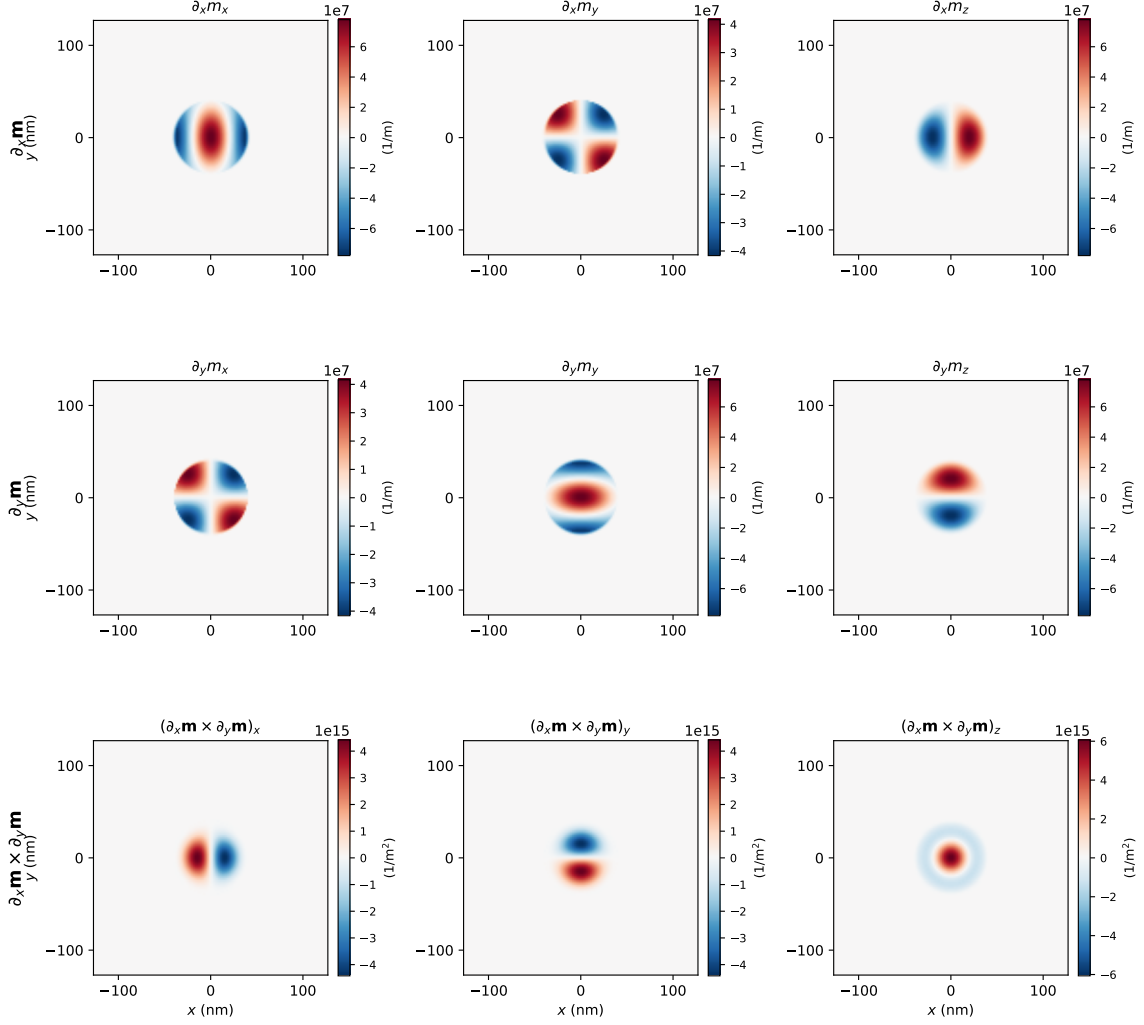


Figure 8: Step-by-step breakdown of the Pontryagin integrand for the same Néel skyrmion as Fig. 7. Top row: the three Cartesian components of $\partial_x \mathbf{m}$ from the central difference (97). Middle row: the same for $\partial_y \mathbf{m}$. Bottom row: components of the cross product $\partial_x \mathbf{m} \times \partial_y \mathbf{m}$. The scalar $\mathbf{m} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m}) = N_{xy}$ that integrates to $4\pi Q$ is shown in Fig. 7(b).

8.1 Discretization

On the lattice with spacing a and periodic indices, the partial derivatives are central differences:

$$(\partial_x \mathbf{m})_{i,j} = \frac{\mathbf{m}_{i,j+1} - \mathbf{m}_{i,j-1}}{2a}, \quad (\partial_y \mathbf{m})_{i,j} = \frac{\mathbf{m}_{i+1,j} - \mathbf{m}_{i-1,j}}{2a}. \quad (97)$$

The Pontryagin density at each site is

$$N_{xy}|_{i,j} = \mathbf{m}_{i,j} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m})_{i,j}, \quad (98)$$

and the integral becomes a Riemann sum over cells of area a^2 :

$$Q = \frac{a^2}{4\pi} \sum_{i=0}^{n_y-1} \sum_{j=0}^{n_x-1} N_{xy}|_{i,j} \quad (99)$$

9 Skyrmion Profile (Initial Condition)

The initial Néel skyrmion profile wraps the 1-D domain wall of Appendix A into a circle of radius R :

$$\theta(r) = 2 \arctan\left(\exp\left(-\frac{r-R}{\Delta}\right)\right), \quad (100)$$

$$m_x = \sin \theta \cos \varphi, \quad m_y = \sin \theta \sin \varphi, \quad m_z = \cos \theta \quad (101)$$

where r is the distance from the skyrmion center, R is the radius of the $m_z = 0$ contour, Δ the wall width, and $\varphi = \text{atan2}(y - y_0, x - x_0)$ is the azimuthal angle (Néel helicity).

The top layer has core down ($m_z = -1$ at center). The bottom layer is its full reversal, $\mathbf{m}_{\text{bot}} = -\mathbf{m}_{\text{top}}$: core up, and in-plane spins pointing inward rather than outward. Two facts make this the right partner. The DMI energy is unchanged under $\mathbf{m} \rightarrow -\mathbf{m}$, so both layers keep the chirality the DMI favours. And the two walls are antiparallel at every point, the ground state of the antiferromagnetic RKKY coupling. Flipping only m_z would give the bottom wall the wrong chirality and leave it parallel to the top wall in-plane, frustrating both couplings.

10 Simulation Parameters

All material parameters are taken from [1] for the optimized SAF stack Pt(3) / Co(1.58) / Ru(0.85) / Pt(0.5) / Co(1.58) / Ru(0.85) (thicknesses in nm). The magnetic thickness used for each Co layer is $t_{\text{Co}} = 1.3$ nm, and the spacer separating the two layers is $d_{\text{Ru}} = 1.35$ nm.

Table 1: Material parameters from [1].

Parameter	Symbol	Value	Unit
Saturation magnetization	M_s	1.43×10^6	A/m
Exchange stiffness	A_{ex}	16×10^{-12}	J/m
DMI constant	D	0.85×10^{-3}	J/m ²
Anisotropy (top layer)	K_{top}	1.3106×10^6	J/m ³
Anisotropy (bottom layer)	K_{bot}	1.31×10^6	J/m ³
Gilbert damping	α	0.14	—
Co layer thickness	t_{Co}	1.3	nm
Spacer thickness	d_{Ru}	1.35	nm
Gyromagnetic ratio	γ	194.8×10^9	rad/(s T)
RKKY coupling field	H_{RKKY}	205	mT
DL-SOT coefficient	χ_{DL}	2.21×10^{-14}	T A ⁻¹ m ²
FL-SOT coefficient	χ_{FL}	0.53×10^{-14}	T A ⁻¹ m ²
Domain wall width	Δ	27	nm

Two values differ from the measured ones on purpose. The paper reports $D = 0.62 \pm 0.24$ mJ/m²; the value 0.85 mJ/m² lies in the upper part of that band. The measured top-layer anisotropy, 1.294×10^6 J/m³ ($\mu_0 H_k = 12.4$ mT), is raised to 1.3106×10^6 J/m³ ($\mu_0 H_k = 36$ mT), as in the paper’s own simulations, so that the background domain is not reversed by the SOT at the largest currents.

Derived quantities

Quantity	Formula	Value
$K_{\text{eff, top}}$	$K_{\text{top}} - \mu_0 M_s^2/2$	$\approx 2.6 \times 10^4 \text{ J/m}^3$
$K_{\text{eff, bot}}$	$K_{\text{bot}} - \mu_0 M_s^2/2$	$\approx 2.5 \times 10^4 \text{ J/m}^3$
Wall-width parameter	$\Delta = \sqrt{A_{\text{ex}}/K_{\text{eff}}}$	$\approx 25 \text{ nm}$
Critical DMI	$D_c = (4/\pi)\sqrt{A_{\text{ex}} K_{\text{eff}}}$	$\approx 0.82 \text{ mJ/m}^2$

Numerical parameters

Parameter	Value	Description
$n_x \times n_y$	256×256	Lattice sites
a	2 nm	Lattice constant
Δt	$5 \times 10^{-14} \text{ s}$	Time step
Relaxation steps	10 000	500 ps at $J = 0$
Drive steps	5 000	250 ps with current
Current density	$J(t)$	current pulse, set per run
Spin polarization	$\hat{\mathbf{p}} = \hat{\mathbf{y}}$	From spin Hall effect
Initial skyrmion radius	93.25 nm	Relaxes to equilibrium
Domain wall width (init.)	27 nm	From paper (Table S2)

Stability condition

The RK4 time step must satisfy:

$$\omega_{\text{max}} \cdot \Delta t < 2.8 \quad (102)$$

where $\omega_{\text{max}} = \gamma \cdot H_{\text{max}}$ and $H_{\text{max}} \approx 4C_{\text{ex}}$ is the largest exchange field on the lattice, reached when neighbouring spins are antiparallel. With $C_{\text{ex}} = 2A_{\text{ex}}/(M_s a^2) \approx 5.6 \text{ T}$, $H_{\text{max}} \approx 22 \text{ T}$ and $\omega_{\text{max}} \cdot \Delta t \approx 0.22$, well within the RK4 stability region.

11 Implementation Notes

Four modelling choices and material-parameter uncertainties limit how closely any simulation of this stack can match experiment.

11.1 Local $N_z = 1$ demag vs full FFT demag

The micromagnetic demagnetization tensor $\hat{N}(\mathbf{k})$ depends on wavevector. For a thin slab of thickness t_{Co} :

$$N_{zz}(k) = f(|k| t_{\text{Co}}), \quad f(x) = \frac{1 - e^{-x}}{x}. \quad (103)$$

- $|k| t_{\text{Co}} \rightarrow 0$ (long wavelength, uniform film): $f \rightarrow 1$, $N_{zz} \rightarrow 1$, $N_{xx,yy} \rightarrow 0$. In real space: $\mathbf{H}_{\text{demag}} = -\mu_0 M_s m_z \hat{\mathbf{z}}$ — a **local** field driven only by m_z at the same point.
- $|k| t_{\text{Co}} \rightarrow \infty$ (short wavelength): $f \rightarrow 0$, no demag.

The **local** $N_z = 1$ **approximation** assumes $N_{zz} = 1$ everywhere and absorbs the demag into the anisotropy:

$$K_{\text{eff}} = K - \frac{1}{2} \mu_0 M_s^2. \quad (104)$$

The main RK4 loop in `src/skyrmion_simulator/simulator/main.py` uses this approximation; it never calls `demag.py`. It is dimensionally correct only when:

1. the texture varies slowly compared to t_{Co} — true here ($R \sim 80$ nm vs $t_{\text{Co}} = 1.3$ nm, $|k|t_{\text{Co}} \sim 0.016$);
2. the magnetization is dominantly out-of-plane.

The second assumption **fails inside the domain wall**, where $\mathbf{m}$ is in-plane, creates surface magnetic charges, and induces non-local stray fields between in-plane spins.

The full FFT demag (in `src/skyrmion_simulator/simulator/demag.py`) computes $\hat{N}(\mathbf{k})$ exactly via FFT, including in-plane components and the inter-layer dipolar kernel across the Ru spacer. MuMax3 does this by default; the simulator's main loop does not. The difference matters for the skyrmion size: non-local demag lowers the in-plane DW energy, allowing a wider wall and a larger skyrmion.

11.2 DMI uncertainty: $D \pm 39\%$

Interfacial DMI cannot be measured directly. Standard techniques:

- **Brillouin light scattering (BLS)**: asymmetric spin-wave dispersion $\omega(+k) \neq \omega(-k)$ caused by DMI. Sensitive to interface roughness and Pt thickness.
- **Domain-wall creep velocity vs in-plane field**: $v(H_x)$ asymmetry yields D .
- **Skyrmion size vs out-of-plane field**: equilibrium R is monotone in D/D_c .

Each method has 10–30% systematic uncertainty (interface quality, sample variation, fitting models). Cross-method agreement is often factor-of-2. Pham *et al.* [1] quote $D = 0.62 \pm 0.24$ mJ/m² ($\approx \pm 39\%$).

The skyrmion size depends strongly on D : near the Bogdanov–Hubert threshold $D_c = (4/\pi)\sqrt{A_{\text{ex}}K_{\text{eff}}}$, the equilibrium R diverges. A $\pm 39\%$ variation in D shifts D/D_c enough to change R by tens of nm.

11.3 Anisotropy uncertainty: $K \pm 7\%$

K is measured via FMR resonance fields, anomalous-Hall hysteresis, or MOKE saturation fields. These methods are well-standardized and agree within 5–10% between labs. Hence $\pm 7\%$ on K .

The catch is that $K_{\text{eff}} = K - \mu_0 M_s^2/2$ is a **small difference of large numbers**:

$$K = 1.294 \times 10^6, \quad \mu_0 M_s^2/2 = 1.285 \times 10^6 \text{ J/m}^3 \implies K_{\text{eff}} \approx 9 \times 10^3 \text{ J/m}^3. \quad (105)$$

A 7% error on K ($\sim 9 \times 10^4$ J/m³) is **10× larger than K_{eff} itself** (and still 3.5× larger for the $K_{\text{eff}} \approx 2.6 \times 10^4$ J/m³ of the 36 mT anisotropy used here) — the system sits in the spin reorientation transition regime where K_{eff} is barely positive. The domain-wall width $\Delta = \sqrt{A_{\text{ex}}/K_{\text{eff}}}$ depends on $1/\sqrt{K_{\text{eff}}}$, so few-percent errors on K propagate to large errors on Δ and the skyrmion size.

11.4 Gilbert damping uncertainty: $\alpha \pm 0.04$

Damping in Co/Pt is enhanced by **spin pumping**: precessing magnetization in Co injects a transverse spin current into Pt, where strong spin-orbit scattering dissipates it. This adds an interfacial term to the bulk Co damping. Measurements:

- **FMR linewidth ΔH** vs frequency f : slope = $\alpha \cdot 2\pi/\gamma$.
- **Time-resolved MOKE** precession decay.

Both give $\alpha \approx 0.14 \pm 0.04$ for the Co/Pt stack ($\sim \pm 30\%$).

In the SAF, the Magnus term cancels between layers (Sec. 7), so the Thiele equation reduces to a damping-limited drift:

$$v \propto \frac{1}{\alpha}. \quad (106)$$

A ± 0.04 uncertainty in α around 0.14 propagates to $\pm 30\%$ in velocity.

11.5 Summary

Approximation	Mechanism	Affects	Magnitude
Local $N_z = 1$ demag	non-local in-plane dipolar ignored	DW energy, skyrmion size	few-% field, larger on size
$D \pm 39\%$	DMI hard to measure	skyrmion size, D_c	exponential near threshold
$K \pm 7\%$	material spread	K_{eff} via subtraction	amplified $\sim 10\times$ on K_{eff}
$\alpha \pm 0.04$	spin pumping enhancement	drift velocity $\propto 1/\alpha$	$\sim 30\%$ on velocity

11.6 Time Integration

The explicit LLGS equation (77) is integrated using the classical 4th-order Runge–Kutta (RK4) method, whose Butcher tableau is

$$\begin{array}{c|ccc}
 0 & & & \\
 \frac{1}{2} & \frac{1}{2} & & \\
 \frac{1}{2} & 0 & \frac{1}{2} & \\
 1 & 0 & 0 & 1 \\
 \hline
 & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6}
 \end{array} \tag{107}$$

At each substep, the spin configuration is renormalized to enforce $|\mathbf{m}| = 1$:

$$\mathbf{k}_1 = \Delta t f(\mathbf{m}) \tag{108}$$

$$\mathbf{k}_2 = \Delta t f\left(\text{norm}\left(\mathbf{m} + \frac{\mathbf{k}_1}{2}\right)\right) \tag{109}$$

$$\mathbf{k}_3 = \Delta t f\left(\text{norm}\left(\mathbf{m} + \frac{\mathbf{k}_2}{2}\right)\right) \tag{110}$$

$$\mathbf{k}_4 = \Delta t f(\text{norm}(\mathbf{m} + \mathbf{k}_3)) \tag{111}$$

$$\mathbf{m}^{n+1} = \text{norm}\left(\mathbf{m}^n + \frac{\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4}{6}\right) \tag{112}$$

Both SAF layers are evolved simultaneously, as the RKKY coupling makes them interdependent. At finite temperature the stochastic equation is integrated with the Heun scheme of Section 5.2 instead.

11.7 Boundary Conditions

Periodic boundary conditions (PBC) are applied in both x and y directions by default, implemented via `np.roll` on the spin arrays. Finite samples use free edges instead: the demag is zero-padded across the free direction (Sec. 4.3.3), and the exchange and DMI stencils see the Rohart–Thiaville free-edge condition $\partial_n \mathbf{m} = (D/2A_{\text{ex}})(\hat{\mathbf{n}} \times \hat{\mathbf{z}}) \times \mathbf{m}$ at the boundary (Sec. 12.2).

11.8 Output Format

Spin configurations are written in LAMMPS dump format. Since OVITO does not natively support spin degrees of freedom, the three spin components (m_x , m_y , m_z) are written as force components (`fx`, `fy`, `fz`), which OVITO can render as vector arrows. Positions are output in nanometers.

Type 1 corresponds to the top Co layer ($z = 0$), type 2 to the bottom Co layer ($z = -t_{\text{Co}}$).

12 Relaxation Convergence and Finite-Size Independence

Before any finite-temperature or current-driven measurement, the SAF skyrmion ($D = 0.85 \text{ mJ/m}^2$, $\mu_0 H_{k,\text{top}} = 12.4 \text{ mT}$) is relaxed to its zero-current ground state. Two questions decide whether a relaxed configuration is a usable reference: (i) how many integration steps are needed for the skyrmion *size* to converge, and (ii) whether the converged size depends on the simulation box. Both are answered by relaxing the same initial skyrmion in four boxes and tracking the size, energy, and residual torque as a function of step.

Setup. $D = 0.85 \text{ mJ/m}^2$, $\mu_0 H_{k,\text{top}} = 12.4 \text{ mT}$, single magnetisation pulse switched off ($J = 0$), overdamped relaxation ($\alpha = 1$, $dt = 5 \times 10^{-14} \text{ s}$) under the periodic (circular-convolution) Newell demag evaluated with the Gauss–Legendre quadrature tensor. Each box is relaxed for 2.5×10^5 steps in chunks of 2000, *never* early-stopping, so the full convergence transient is recorded. After every chunk the relaxed top-layer state is reduced to: the largest-connected-component (LCC) ellipse axes D_1, D_2 (PBC-aware second moments), the total energy E , and the percentiles of the per-site tangential torque $|\mathbf{m} \times (\mathbf{m} \times \mathbf{H})|$ (maximum, 99th percentile, median, RMS).

grid	size (nm)	steps	max $ \tau $	τ_{p99}	τ_{med}	$D_1 = D_2$ (nm)	E (J)
256×256	512×512	250,000	5.82×10^{-5}	5.77×10^{-5}	1.21×10^{-6}	206.5	-1.11×10^{-16}
350×350	700×700	250,000	5.77×10^{-5}	5.68×10^{-5}	4.60×10^{-8}	206.5	-2.09×10^{-16}
350×500	700×1000	250,000	5.76×10^{-5}	5.64×10^{-5}	5.89×10^{-9}	206.5	-2.98×10^{-16}
500×500	1000×1000	250,000	5.76×10^{-5}	5.59×10^{-5}	1.34×10^{-9}	206.5	-4.26×10^{-16}

Table 2: Relaxed-state observables after 2.5×10^5 steps. The relaxed diameter is identical to within the LCC-ellipse resolution across all boxes; the total energy scales with the ferromagnetic background area, as expected for a fixed-size skyrmion in a larger cell.

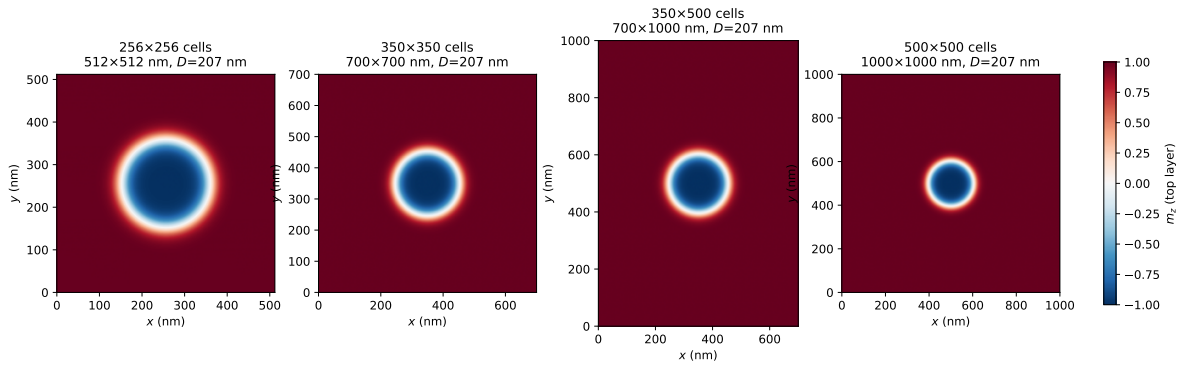


Figure 9: Relaxed top-layer m_z in each box, drawn at true physical scale. The skyrmion ($m_z < 0$ core) is the same $\approx 207 \text{ nm}$ object in every box; only the surrounding ferromagnetic background ($m_z = +1$) grows. The 256^2 box is the tightest confinement and the 500×500 box the most generous, yet the core is unchanged.

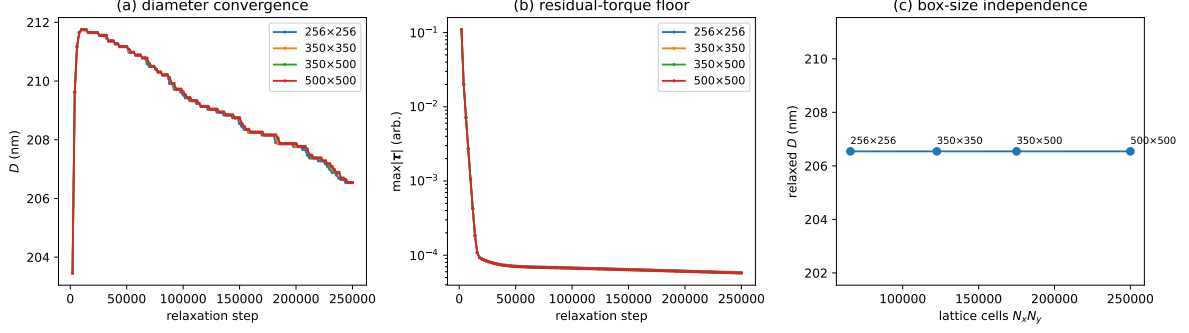


Figure 10: (a) Diameter vs step: the skyrmion overshoots to ≈ 211.7 nm within the first $\sim 10^4$ steps, then relaxes slowly to 206.5 nm; the four boxes overlap. (b) Maximum tangential torque vs step: it falls four orders of magnitude in $\sim 3 \times 10^4$ steps to a flat floor of $\approx 6 \times 10^{-5}$. (c) Relaxed diameter vs lattice-cell count: flat, i.e. box-size independent.

Why 2.5×10^5 steps and not 4×10^4 . Figure 10(a) shows the diameter is *non-monotonic*: it overshoots to ≈ 211.7 nm during the initial collapse, then drifts downward over the next $\sim 2 \times 10^5$ steps to the converged 206.5 nm. Stopping at 4×10^4 steps would report ≈ 210 nm — roughly 1.7% too large and still visibly drifting. The longer run is required precisely because the size, not just the torque, must plateau.

Box-size independence. The relaxed diameter is 206.5 nm in all four boxes (Table 2, Fig. 10(c)), even though the cell area grows by $3.8\times$ from 256^2 to 500×500 . This is the expected behaviour of a *compensated* synthetic antiferromagnet: the two antiparallel layers nearly cancel the net magnetostatic moment, so the periodic images contribute negligibly and the confinement is set by the intrinsic balance of exchange, DMI, and anisotropy rather than by the cell. The practical consequence is that the periodic and the isolated (zero-padded, free-boundary) demag give the same relaxed size for this stack, so the boundary-condition choice is a question of matching the reference paper, not of physics.

Circular at $J = 0$. The relaxed skyrmion has $D_1 = D_2$ to within the ellipse resolution: with no spin-orbit torque there is no preferred in-plane direction and the ground state is axisymmetric. The elliptical ($D_1 \neq D_2$) deformation studied elsewhere in this work appears only once a current drives the skyrmion.

The residual-torque floor is not under-relaxation. The maximum tangential torque does not vanish; it plateaus at $\approx 6 \times 10^{-5}$ (Fig. 10(b)). Two facts identify this as a benign max-norm artifact rather than an unrelaxed skyrmion. First, $\tau_{p99} \approx \max|\tau|$ while the median is three to five orders of magnitude smaller (Table 2): the residual is a broad, low-amplitude field spread across the ferromagnetic background, not a torque spike localised at the domain wall. Second, the floor does *not* grow with lattice size — if anything it is slightly lower on the larger boxes — whereas a genuinely under-relaxed wall would tighten and the peak torque would track the wall, not the background. The skyrmion itself (where the size and energy are determined) is fully relaxed; the floor is the maximum over an ever-larger set of near-zero background sites.

Demag-method robustness. These results use the Gauss–Legendre quadrature evaluation of the Newell tensor. Re-relaxing with the closed-form Newell–Williams–Dunlop tensor (exact to machine precision) yields the identical 206.5 nm diameter, confirming that the relaxed size is insensitive to the demag-evaluation method and that the convergence behaviour above is a property of the physics, not of the quadrature.

12.1 Free top/bottom boundaries: the $D > D_c$ expansion

Repeating the relaxation with free (open) top and bottom edges across the width y — the $2n_y \times n_x$ zero-padded racetrack demag, here with the exchange/DMI still on the y -torus (the

demag-only free- y case of Sec. 12.2) — changes the picture completely. All other settings are unchanged ($D = 0.85$ mJ/m², $J = 0$, $\alpha = 1$, 2.5×10^5 steps).

grid	size (nm)	max $ \tau $	τ_{med}	D_1 (nm)	D_2 (nm)	E (J)
256×256	512×512	1.39×10^{-3}	1.62×10^{-4}	406.4	401.9	-1.15×10^{-16}
350×350	700×700	1.55×10^{-3}	1.63×10^{-4}	436.5	435.8	-2.13×10^{-16}
350×500	700×1000	1.58×10^{-3}	4.55×10^{-5}	436.8	436.2	-3.03×10^{-16}
500×500	1000×1000	1.58×10^{-3}	8.04×10^{-6}	436.7	436.6	-4.31×10^{-16}

Table 3: Relaxed-state observables with free top/bottom boundaries after 2.5×10^5 steps. In contrast to the periodic case (Table 2, a box-independent 206.5 nm), the diameter is roughly doubled and has *not* converged — see Fig. 11.

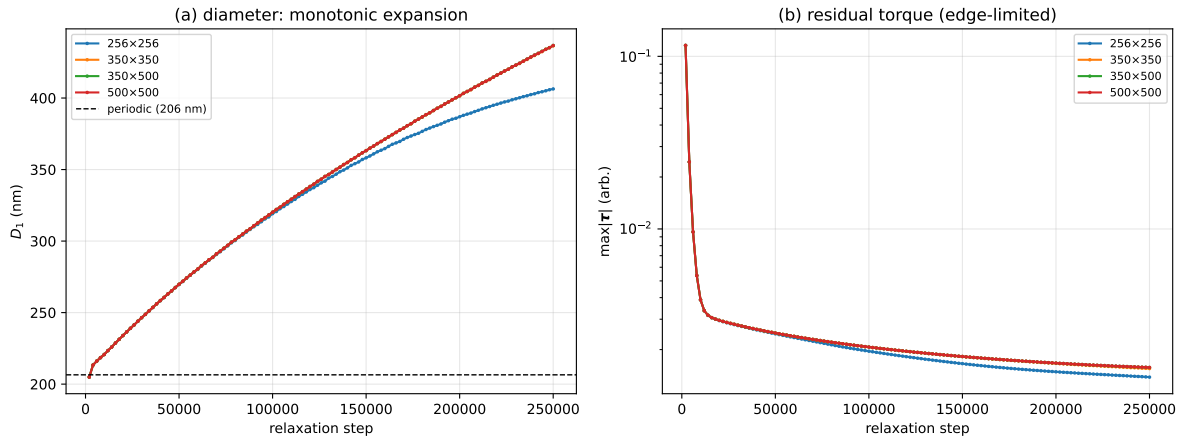


Figure 11: (a) Diameter vs step under free top/bottom boundaries: the skyrmion expands monotonically and is still growing at 2.5×10^5 steps ($\approx +0.7$ nm per 10^3 steps). The three larger boxes share one trajectory because none has yet reached its confining edge; only 256^2 (512 nm) is small enough to bend below the others. The periodic equilibrium (206.5 nm, dashed) sits far below. (b) Maximum torque plateaus at $\approx 1.5 \times 10^{-3}$ ($\sim 25 \times$ the periodic floor) — the persistent torque of the still-moving wall and the open edges.

The skyrmion does not settle. The reported 406–437 nm are mid-expansion snapshots, not equilibria: the diameter is still rising at $+0.7$ nm per 10^3 steps over the last 10^5 steps. The three larger boxes (350^2 , 350×500 , 500^2) report nearly identical sizes (≈ 436 – 437 nm) *not* because they have converged but because they follow the same free-expansion trajectory — at 1.5×10^5 steps all three read 363 nm — and none has yet grown enough to feel its (different) box edge. The 256^2 box is the only one small enough that the skyrmion already touches the edge by $\sim 1.2 \times 10^5$ steps, which is why it bends below the others and its expansion rate halves.

The residual torque is an edge/expansion transient, not bulk under-relaxation. $\max|\tau|$ holds at $\sim 1.5 \times 10^{-3}$ (the moving wall and the open edges), but the *median* torque falls by more than an order of magnitude as the box grows (1.6×10^{-4} at 256^2 to 8.0×10^{-6} at 500^2): the interior is increasingly well-relaxed while the residual concentrates at the expanding wall and the free boundary.

Why this happens. At $D = 0.85$ mJ/m² with $\mu_0 H_{k,\text{top}} = 12.4$ mT the stack sits at $D/D_c = 1.75$ ($\sigma_{\text{DW}} = 4\sqrt{A_{\text{ex}} K_{\text{eff}}} - \pi D = -1.14 \times 10^{-3}$ J/m² < 0 , Sec. 12). With negative wall energy the isolated skyrmion has no finite equilibrium radius — it lowers its energy by enlarging toward a stripe domain. Periodic boundaries on all sides supply a confining image interaction that pins it at the metastable, box-independent 206.5 nm; removing the confinement across the width (free top/bottom) eliminates that restoring term and the skyrmion expands until arrested

by the physical edge. Consistently, the energy grows more negative as the skyrmion (and its surrounding domain) enlarges.

Consequence. For this stack a free-edge — or a true finite-width track — skyrmion size is set by confinement, not by intrinsic energetics, and there is no $T = 0$ free-relaxation equilibrium to use as a seed. An intrinsically stable, finite skyrmion would require either $D < D_c \approx 0.49 \text{ mJ/m}^2$ (the lower end of the paper’s reported $D = 0.62 \pm 0.24 \text{ mJ/m}^2$ band, where $\sigma_{\text{DW}} > 0$) or hard-edge geometric confinement at the intended width.

12.2 Boundary conditions: periodic vs racetrack vs masked band

The periodic bulk, the production racetrack, and the masked-band diagnostic differ only in the top/bottom (y) boundary, and that boundary is applied *identically to both* long-range couplings — magnetostatics and exchange/DMI. Fig. 12 contrasts the three. The x direction (track length) is periodic throughout.

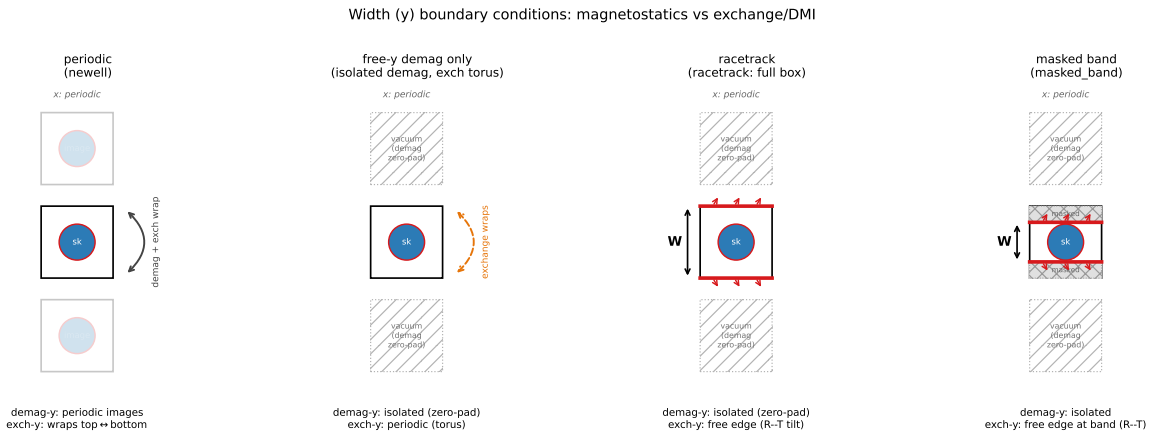


Figure 12: Width (y) boundary conditions. **Periodic (newell)**: both the demag kernel and the exchange/DMI stencil wrap top $\leftrightarrow$ bottom, so the cell sees an infinite stack of y images and behaves as a 2-torus. **Free- y demag only**: the demag kernel is built on the $2n_y \times n_x$ zero-padded grid (isolated across y) but the exchange/DMI stencil still wraps — the short-range torques remain a torus; not a production BC, shown to motivate why both couplings must be freed (Sec. 12.1 uses this case). **Racetrack (racetrack)**: the full box is the track — the demag kernel is zero-padded across y (no images) *and* the exchange/DMI stencil is cut at the box edge with the Rohart–Thiaville free-edge tilt (red arrows); periodic x , free y , no mask, track width $L_y = n_y a$. **Masked band (masked_band)**: the same racetrack demag, but the magnetic region is a centred band of width $W < L_y$ with vacuum margins carrying the free edge.

Periodic (newell). The demag is a circular convolution on the $n_y \times n_x$ grid and the exchange/DMI stencil wraps $y \pm 1$ across the box: the slab feels an infinite array of y images, a 2-torus on both couplings.

Racetrack (racetrack). The full box is the track. The demag kernel is built on the doubled $2n_y \times n_x$ zero-padded grid, a linear convolution isolating a single finite-width stripe (no y images); and the exchange/DMI stencil is cut at the box edge (rows 0 and $n_y - 1$), where ghost cells carry the Rohart–Thiaville free-edge condition $\partial_n \mathbf{m} = (D/2A_{\text{ex}}) (\hat{\mathbf{n}} \times \hat{\mathbf{z}}) \times \mathbf{m}$. Periodic x , free y , no mask: the track width is the transverse box extent $L_y = n_y a$.

Masked band (masked_band). The same racetrack demag, but the magnetic region is a centred band of width $W < L_y$; vacuum margins above and below carry the free edge. This imposes a track width narrower than the box and is exercised by the diagnostic `validate_relax_torque masked_band`.

Both couplings must be free. Isolating only the demag (free demag, periodic exchange) leaves the short-range torques on a torus and, for $D > D_c$, the texture still expands without bound (Sec. 12.1). The complete racetrack frees both. The long-range demag dominates the field change (here $\approx 17\%$ versus periodic), while the exchange/DMI edge is local — it matters once the texture reaches the boundary (e.g. under the SOT Hall drift), not for a centred skyrmion. The C++ and Python implementations are identical, cross-validated to FFT round-off ($\sim 10^{-12}$).

12.3 DMI sweep: dialing a stable, larger racetrack skyrmion

The racetrack relaxation at $D = 0.47 \text{ mJ/m}^2$ settles a compact $\approx 90 \text{ nm}$ skyrmion. To reach the paper’s $\approx 180\text{--}200 \text{ nm}$ scale while staying intrinsically stable, we sweep D across the reported band (0.62 ± 0.24 , i.e. $0.38\text{--}0.86 \text{ mJ/m}^2$) and relax the full-box racetrack BC (Sec. 12.2) on two boxes (256^2 , 350^2 , $J = 0$, $\alpha = 1$, 2.5×10^5 steps), reading the relaxed LCC ellipse D_1, D_2 and the box-to-box consistency.

D (mJ/m ²)	D_1/D_2 (256 ²)	D_1/D_2 (350 ²)	D_1 (nm) (350 ²)	$ \Delta D_1 /D_1$ (box)	class
0.47	1.00	1.00	91.2	0.012	compact
0.52	1.00	1.00	144.4	0.018	compact
0.57	1.01	1.00	209.1	0.020	compact
0.62	1.04	1.01	270.5	0.017	compact (edge)
0.66	1.07	1.01	314.2	0.013	compact (edge)

Table 4: Relaxed racetrack skyrmion versus DMI. D_1/D_2 is the LCC ellipse aspect ratio per box; D_1 is the major axis on the 350^2 box; $|\Delta D_1|/D_1$ is the relative difference of D_1 between the two boxes (box-independence test).

All five are stable, compact skyrmions — none is a stripe. The decisive test is box-independence: a $D > D_c$ stripe has no finite equilibrium and grows with the box (as in Sec. 12.1, D_1 rising from 406 to 437 nm with box size). Here every D gives $|\Delta D_1|/D_1 \leq 2\%$ across the two boxes — a genuine finite-radius equilibrium. The size grows monotonically $90 \rightarrow 314 \text{ nm}$ as $D : 0.47 \rightarrow 0.66$, consistent with the equilibrium radius diverging as $D \rightarrow D_c^-$.

The high- D ellipticity is a finite-box edge effect, not an instability. D_1/D_2 is ≈ 1.00 up to 0.57; at 0.62–0.66 it rises to 1.04–1.07 *only on the 256^2 box* and stays ≈ 1.01 on 350^2 . A $\approx 310 \text{ nm}$ skyrmion in a 512 nm (256-cell) box sits only $\sim 100 \text{ nm}$ from each free top/bottom edge, so it elongates along the periodic x ; on the 700 nm box it is round. The effect vanishes with box size, the signature of confinement rather than intrinsic anisotropy.

The relaxation does not stop on the torque gate, and should not. Every entry runs the full 2.5×10^5 steps without triggering the 10^{-5} torque tolerance: the racetrack $\max|\tau|$ floor is the max-norm artifact of Sec. 12 (a well-relaxed skyrmion with residual torque delocalised in the ferromagnetic background), not under-relaxation. Size and box-independence, not the torque gate, are the stability criteria.

Target. The $180\text{--}200 \text{ nm}$ goal corresponds to $D \approx 0.57$ (209 nm) or, interpolating $0.52 \rightarrow 0.57$, $D \approx 0.545$ ($\approx 185 \text{ nm}$) — both well inside the paper’s band and comfortably below the effective D_c .

Confirmation at $D = 0.545 \text{ mJ/m}^2$. A dedicated relaxation at the chosen $D = 0.545$ settles a compact skyrmion of $D_1 = 184.5 \text{ nm}$ on the production 350×500 box and $180.7/180.2 \text{ nm}$ on 256^2 — box independent to 2% and on target. Figures 13 and 14 show the relaxed configuration and the diameter / torque evolution.

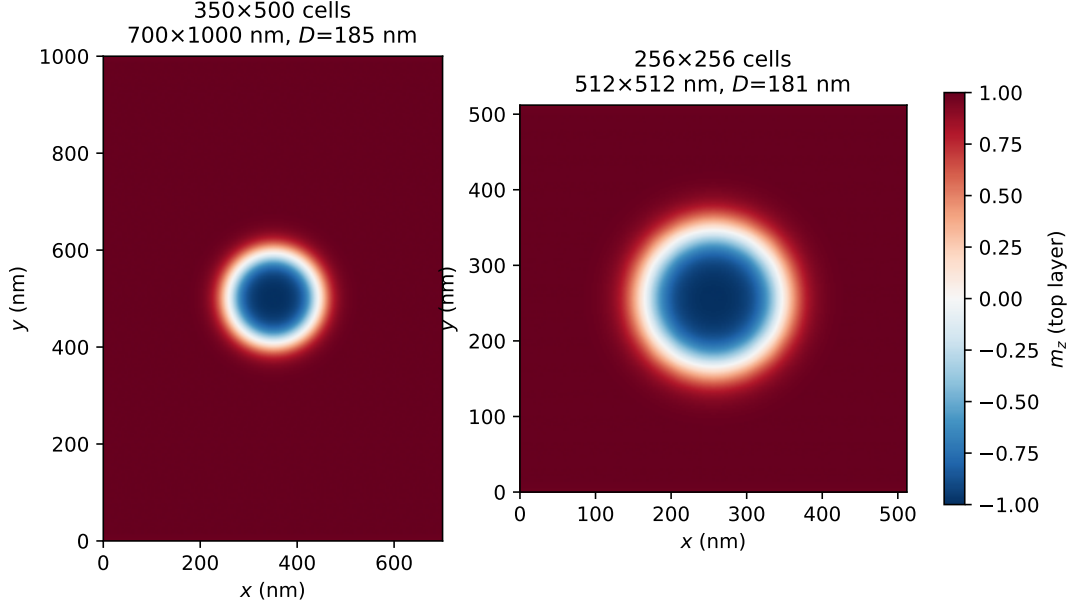


Figure 13: Relaxed top-layer m_z at $D = 0.545$ mJ/m² (racetrack BC, $J = 0$) for the two boxes finished so far. The compact ≈ 182 nm skyrmion is centred and round on both, the target track scale.

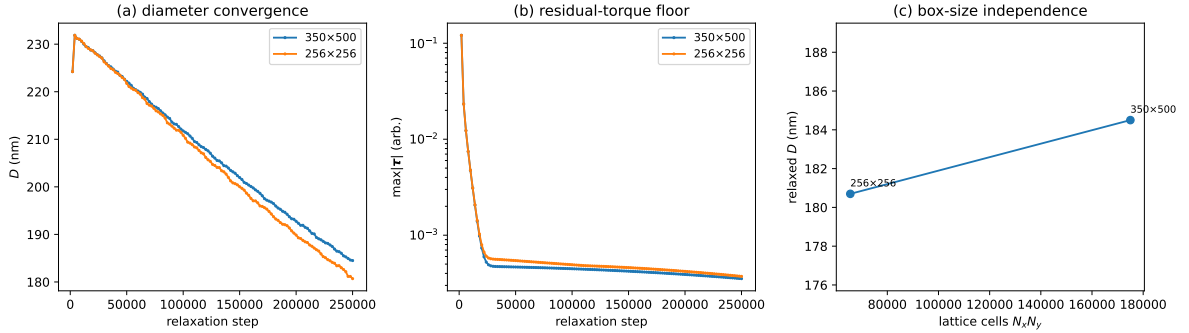


Figure 14: (a) Diameter vs step: overshoot then relaxation to a flat, box-independent ≈ 182 – 185 nm plateau. (b) Maximum tangential torque vs step: falls to the max-norm floor (Sec. 12), not the 10^{-5} gate. (c) Relaxed diameter vs lattice cells: flat — the equilibrium is set by D , not the box.

A Derivation of the Domain Wall Profile

The skyrmion initial condition uses the one-dimensional domain wall profile obtained from the Euler–Lagrange equation. This appendix derives that profile from first principles.

From 3D energy to 1D energy density

The starting point is the three-dimensional micromagnetic energy for a thin film with exchange and effective anisotropy (Sections 2.1 and 2.3–2.4):

$$E = \int_V \left[A_{\text{ex}} |\nabla \mathbf{m}|^2 - K_{\text{eff}} m_z^2 \right] dV \quad (113)$$

For a domain wall that varies only along x in a film of thickness t_{Co} and cross-sectional area $L_y t_{\text{Co}}$, the volume integral reduces to:

$$E = L_y t_{\text{Co}} \int_{-\infty}^{+\infty} \left[A_{\text{ex}} \left| \frac{d\mathbf{m}}{dx} \right|^2 - K_{\text{eff}} m_z^2 \right] dx \quad (114)$$

We parametrize the unit magnetization by the polar angle $\theta(x)$ and azimuthal angle $\phi(x)$:

$$\mathbf{m} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \quad (115)$$

Computing the gradient magnitude:

$$\left| \frac{d\mathbf{m}}{dx} \right|^2 = \left(\frac{d\theta}{dx} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{dx} \right)^2 \quad (116)$$

For a Néel wall (or Bloch wall), the azimuthal angle is constant ($d\phi/dx = 0$), so the second term vanishes. Substituting $m_z = \cos \theta$ and noting that $-K_{\text{eff}} \cos^2 \theta = -K_{\text{eff}} + K_{\text{eff}} \sin^2 \theta$ (dropping the constant), the energy per unit cross-section becomes:

$$\frac{E}{L_y t_{\text{Co}}} = \int_{-\infty}^{+\infty} \varepsilon(x) dx \quad (117)$$

with the one-dimensional energy density:

$$\boxed{\varepsilon = A_{\text{ex}} \left(\frac{d\theta}{dx} \right)^2 + K_{\text{eff}} \sin^2 \theta} \quad (118)$$

The first term is the exchange cost of spatial variations in θ ; the second is the anisotropy cost for deviating from the easy-axis directions $\theta = 0$ ($m_z = +1$) or $\theta = \pi$ ($m_z = -1$).

Euler–Lagrange equation

Minimizing the total energy $E = \int_{-\infty}^{+\infty} \varepsilon dx$ with respect to $\theta(x)$ gives the Euler–Lagrange equation:

$$2A_{\text{ex}} \frac{d^2 \theta}{dx^2} = K_{\text{eff}} \sin(2\theta) \quad (119)$$

First integral

Multiplying both sides by $d\theta/dx$ and integrating once yields a first integral (conservation of “energy” in the mechanical analogy):

$$A_{\text{ex}} \left(\frac{d\theta}{dx} \right)^2 = K_{\text{eff}} \sin^2 \theta \quad (120)$$

which gives:

$$\frac{d\theta}{dx} = -\frac{\sin \theta}{\Delta}, \quad \Delta \equiv \sqrt{\frac{A_{\text{ex}}}{K_{\text{eff}}}} \quad (121)$$

The negative sign ensures θ decreases from π to 0 as x increases (wall centered at $x = 0$).

Exact solution

Separating variables in (121):

$$\int \frac{d\theta}{\sin \theta} = -\frac{x}{\Delta} + C \quad (122)$$

Using the identity $\int \csc \theta d\theta = \ln |\tan(\theta/2)|$:

$$\ln \left| \tan \frac{\theta}{2} \right| = -\frac{x}{\Delta} \quad (123)$$

where the integration constant is fixed by requiring $\theta(0) = \pi/2$ (wall center). Solving for θ :

$$\boxed{\theta(x) = 2 \arctan \left(\exp \left(-\frac{x}{\Delta} \right) \right)} \quad (124)$$

Properties

The solution (124) has the following limits:

- $x \rightarrow -\infty$: $\theta \rightarrow \pi$ ($m_z = -1$, down domain)
- $x = 0$: $\theta = \pi/2$ ($m_z = 0$, wall center)
- $x \rightarrow +\infty$: $\theta \rightarrow 0$ ($m_z = +1$, up domain)

The magnetization component $m_z = \cos \theta$ transitions smoothly as:

$$m_z(x) = -\tanh \left(\frac{x}{\Delta} \right) \quad (125)$$

confirming that Δ is the characteristic wall width. Using the parameters from [1], $\Delta = \sqrt{A_{\text{ex}}/K_{\text{eff}}} \approx 27$ nm.

Extension to a skyrmion

For a radially symmetric skyrmion centered at the origin, the full 2D variational problem has no closed-form solution because the DMI and the curvature of the domain wall add additional terms. However, the radial profile $\theta(r)$ is well approximated by wrapping the 1D wall into a circle at the skyrmion radius R_{sk} :

$$\theta(r) = 2 \arctan \left(\exp \left(-\frac{r - R_{sk}}{\Delta} \right) \right) \quad (126)$$

This gives:

- $r \ll R_{sk}$: $\theta \rightarrow \pi$ (core, $m_z = -1$)
- $r = R_{sk}$: $\theta = \pi/2$ ($m_z = 0$ contour)
- $r \gg R_{sk}$: $\theta \rightarrow 0$ (background, $m_z = +1$)

The skyrmion radius R_{sk} is not a free parameter — it is determined by the balance of exchange, DMI, and anisotropy energies. In MuMax3, this equilibrium is found numerically: an arbitrary initial guess is evolved under the LLG equation with high damping until the system relaxes to the energy minimum. Our simulator performs the same relaxation during the $J = 0$ phase. The profile (126) simply provides a physically motivated initial condition that is already close to equilibrium, minimizing the relaxation time needed.

B The Constraint $\mathbf{m} \cdot d\mathbf{m}/dt = 0$

The identity $\mathbf{m} \cdot d\mathbf{m}/dt = 0$ is used throughout Section 5.1 (and is the geometric reason the LLGS equation preserves $|\mathbf{m}| = 1$). It follows from two steps.

Step 1: algebraic identity (always true)

By the product rule applied to $|\mathbf{m}|^2 = \mathbf{m} \cdot \mathbf{m}$:

$$\frac{d|\mathbf{m}|^2}{dt} = \frac{d(\mathbf{m} \cdot \mathbf{m})}{dt} = \frac{d\mathbf{m}}{dt} \cdot \mathbf{m} + \mathbf{m} \cdot \frac{d\mathbf{m}}{dt} = 2\mathbf{m} \cdot \frac{d\mathbf{m}}{dt}. \quad (127)$$

Hence

$$\mathbf{m} \cdot \frac{d\mathbf{m}}{dt} = \frac{1}{2} \frac{d|\mathbf{m}|^2}{dt}, \quad (128)$$

which holds for *any* vector $\mathbf{m}(t)$, with or without the unit-norm constraint.

Step 2: apply the constraint $|\mathbf{m}| = 1$

If $|\mathbf{m}| = 1$ at all times then $|\mathbf{m}|^2 = 1$ is a constant, so

$$\frac{d|\mathbf{m}|^2}{dt} = \frac{d(1)}{dt} = 0. \quad (129)$$

Substituting into (128):

$$\mathbf{m} \cdot \frac{d\mathbf{m}}{dt} = 0.$$

(130)

Geometric meaning

The state vector $\mathbf{m}$ lives on the unit two-sphere $S^2 \subset \mathbb{R}^3$. The tangent space at any point $\mathbf{m} \in S^2$ is the plane perpendicular to the radial direction $\mathbf{m}$ itself. So any velocity $d\mathbf{m}/dt$ that keeps $\mathbf{m}$ on S^2 must be perpendicular to $\mathbf{m}$, which is precisely statement (130).

Sphere S^2	Tangent plane at $\mathbf{m}$
$ \mathbf{m} = 1$	$d\mathbf{m}/dt \perp \mathbf{m}$

Why the LLGS RHS preserves the constraint

Every term in the LLGS equation (77) has the form $\mathbf{m} \times (\dots)$ — precession, damping, and both SOT torques are cross products with $\mathbf{m}$. A cross product $\mathbf{m} \times \mathbf{X}$ is orthogonal to both factors, so for any vector $\mathbf{X}$:

$$\mathbf{m} \cdot (\mathbf{m} \times \mathbf{X}) = 0. \quad (131)$$

Therefore dotting the entire LLGS RHS with $\mathbf{m}$ gives zero, $\mathbf{m} \cdot d\mathbf{m}/dt = 0$ is enforced by the equation of motion, and the unit-norm constraint $|\mathbf{m}| = 1$ is conserved *exactly* by the continuous-time dynamics. The RK4 integrator introduces a small $\mathcal{O}(\Delta t^5)$ drift that is removed by an explicit renormalization step $\mathbf{m} \leftarrow \mathbf{m}/|\mathbf{m}|$ at the end of each step (see `normalize()` in `src/skymion_simulator/simulator/integrator.py`).

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